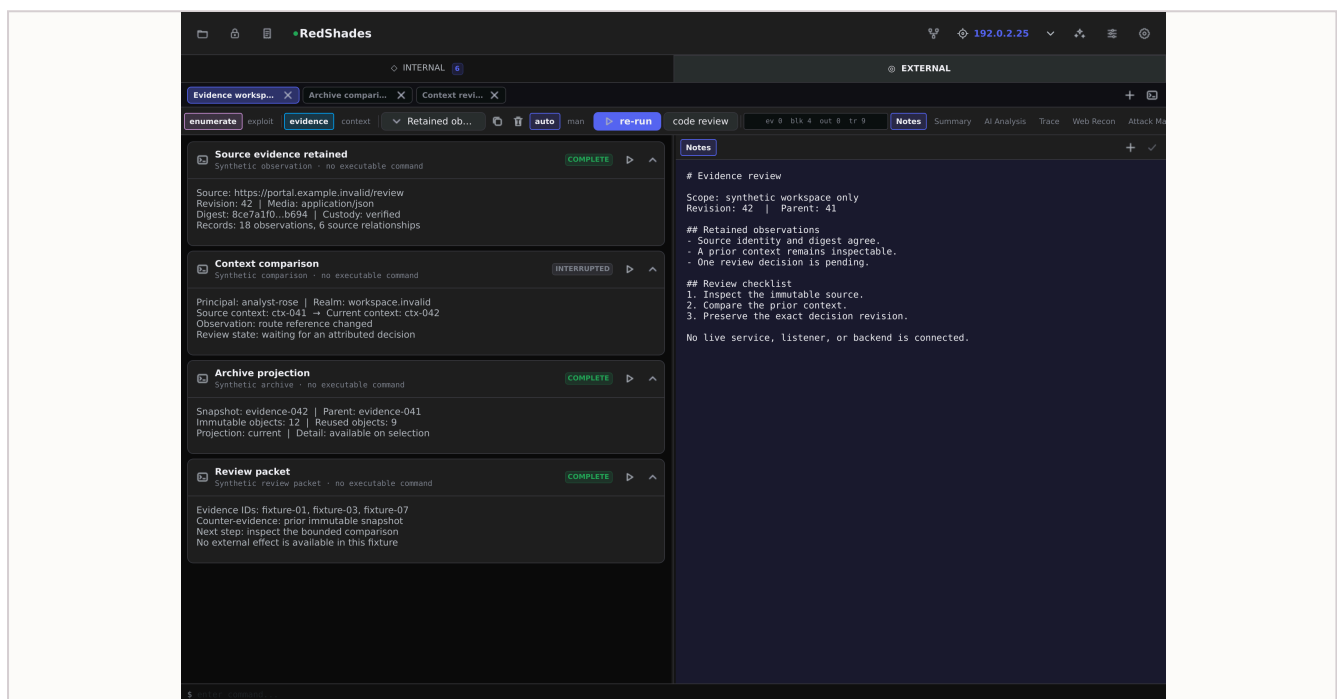


PLATFORM SCOPE

RedShades orchestrates evidence-driven security assessments.

RedShades coordinates security assessments through 88 conditional stages in the Validated Core. Advanced extends the same evidence and routing model across ten specialist assessment domains. Ultimate adds defensive patching and forensic workflows, bringing the catalog to 198 stages, including 13 C2 continuation stages.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

A native Qt workspace connects findings to their sources and keeps investigation context available across sessions. Capability generation and runtime repair support work that encounters missing interactions or software defects.

Implemented

Reference 01

authoritative engagement architecture

EVIDENCE-DRIVEN SCHEDULING

The scheduler selects work from current evidence

The scheduler evaluates 88 automatic stage declarations against current evidence. Independent branches advance when their inputs are ready; loops revisit useful frontiers as observations change. The Executive maintains the wider view while ACP supplies missing capabilities and AEP repairs the underlying software.

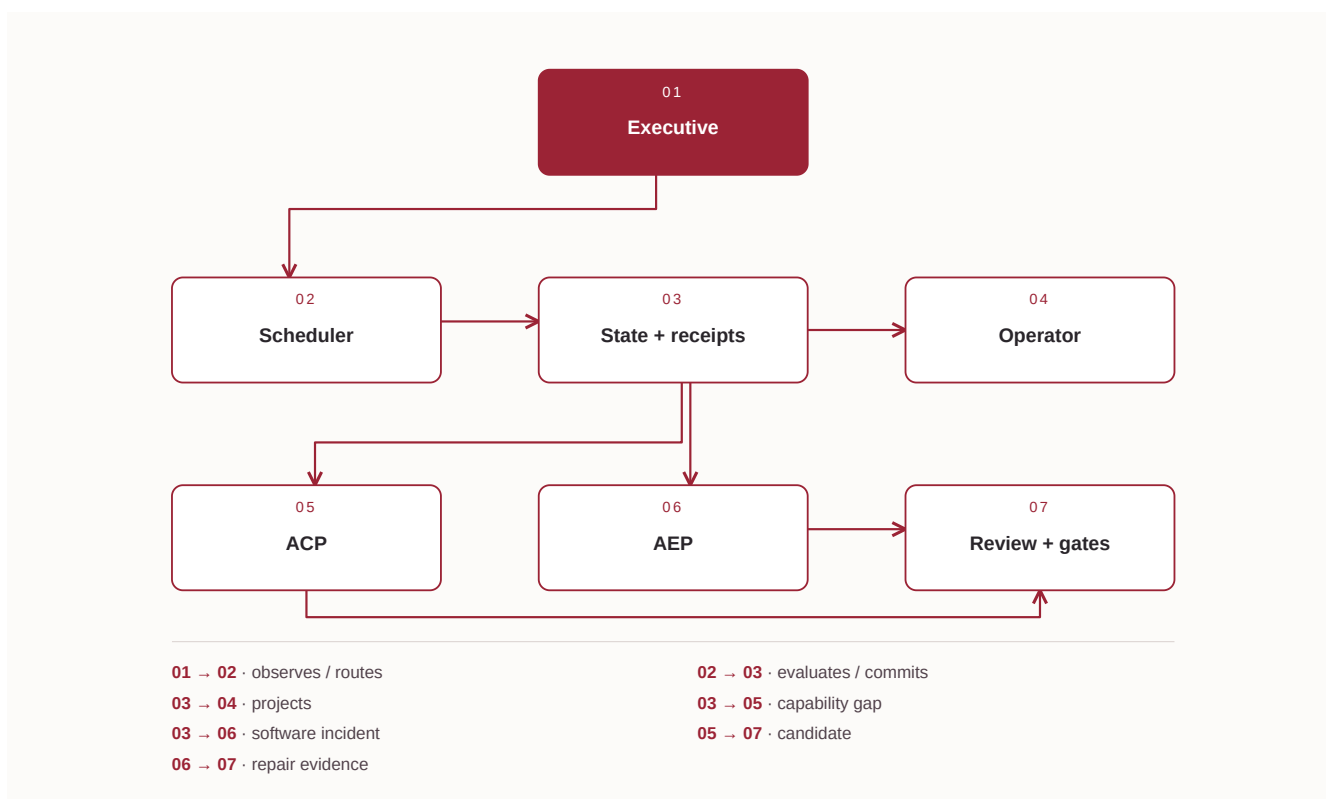


Diagram of the labeled component relationships.

Scheduling, supervision, model interpretation, capability generation and platform repair have distinct responsibilities. Shared records preserve their results across stage boundaries and model sessions.

Implemented

Reference 02
subsystem glossary

CHILD INVESTIGATIONS

Child investigations retain their parent context

A newly available route can open a child subchain with its own scope and execution context. Parent and child retain their relationship, so a deeper investigation remains part of the same attributed engagement.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

Frontiers track eligible and waiting work. Branch-local decisions allow unrelated work to continue; parent completion accounts for attached investigations. This is a graph of evolving work rather than a fixed checklist.

Implemented

Reference 03

authoritative engagement architecture

MODEL ALLOCATION

Deterministic control surrounds model decisions

RedShades describes its design philosophy as 70% deterministic and 30% neural. Software controls scheduling, state, policy and validation. Models interpret ambiguity and contribute to capability development and engineering work.

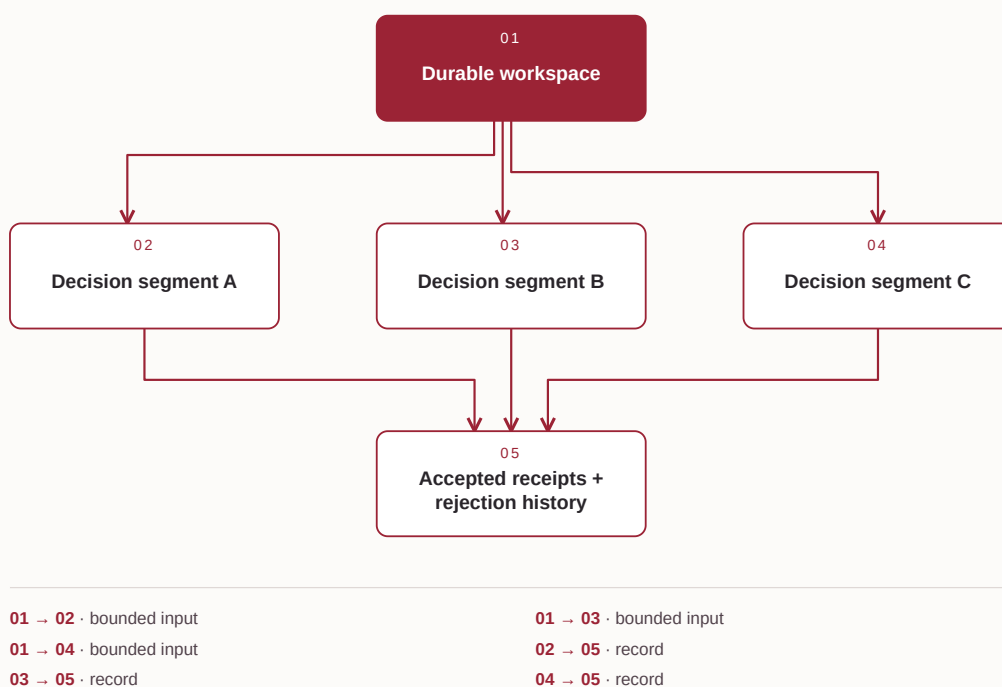


Diagram of the labeled component relationships.

The 70/30 ratio expresses the design philosophy rather than a measured code, runtime or success percentage. Bounded decisions use specialist calls; longer engineering tasks retain persistent sessions and their supporting evidence.

Design philosophy

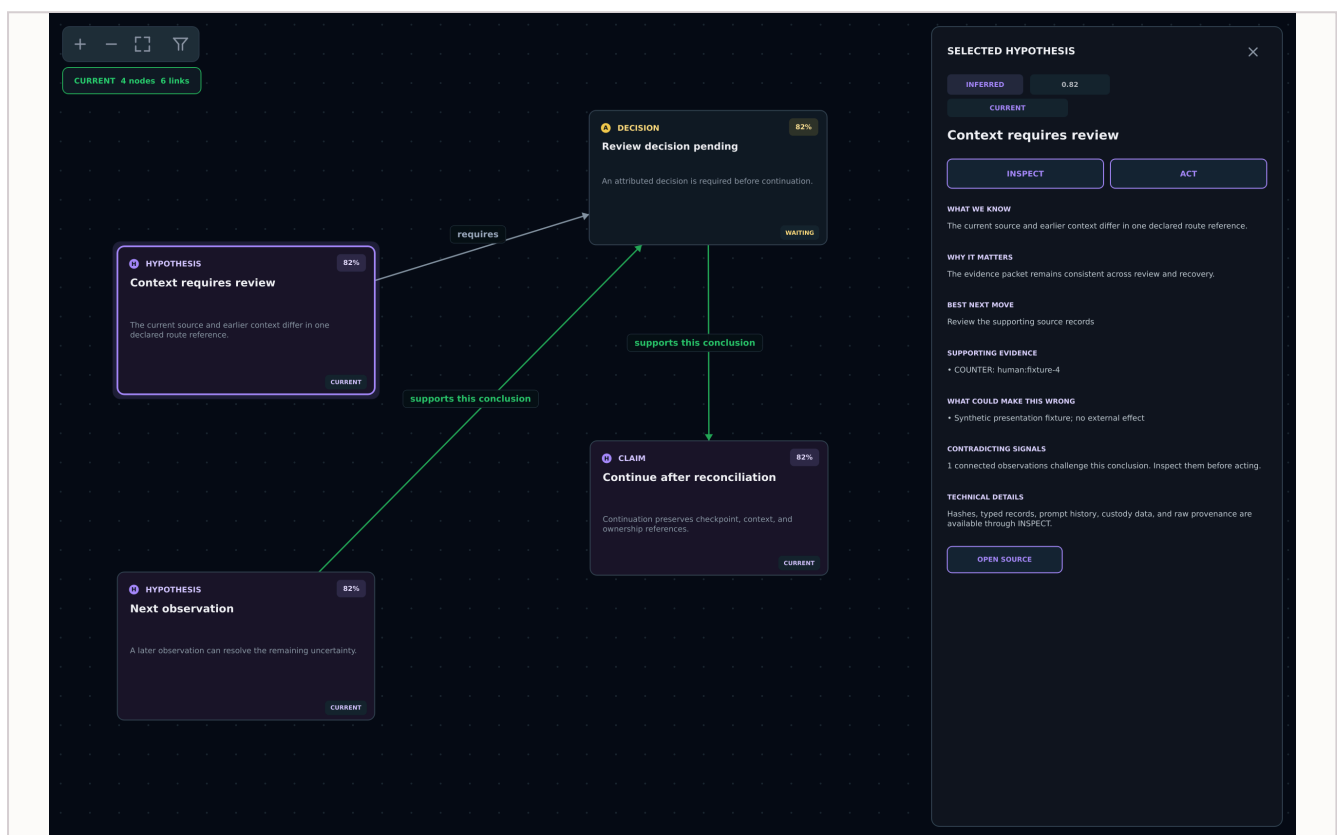
Reference 04

acp vertical session

SHARED ENGAGEMENT STATE

An engagement memory shared by every subsystem

Evidence, identities, routes, artifacts, capabilities and outcomes are linked in versioned engagement records. The scheduler, model workspaces and operator views can work from the same account of the investigation.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

Revision-bound records preserve the inputs behind each decision. Canonical state supplies the architecture for coordinated work across tools and sessions; source references document the implementation boundaries.

Implemented

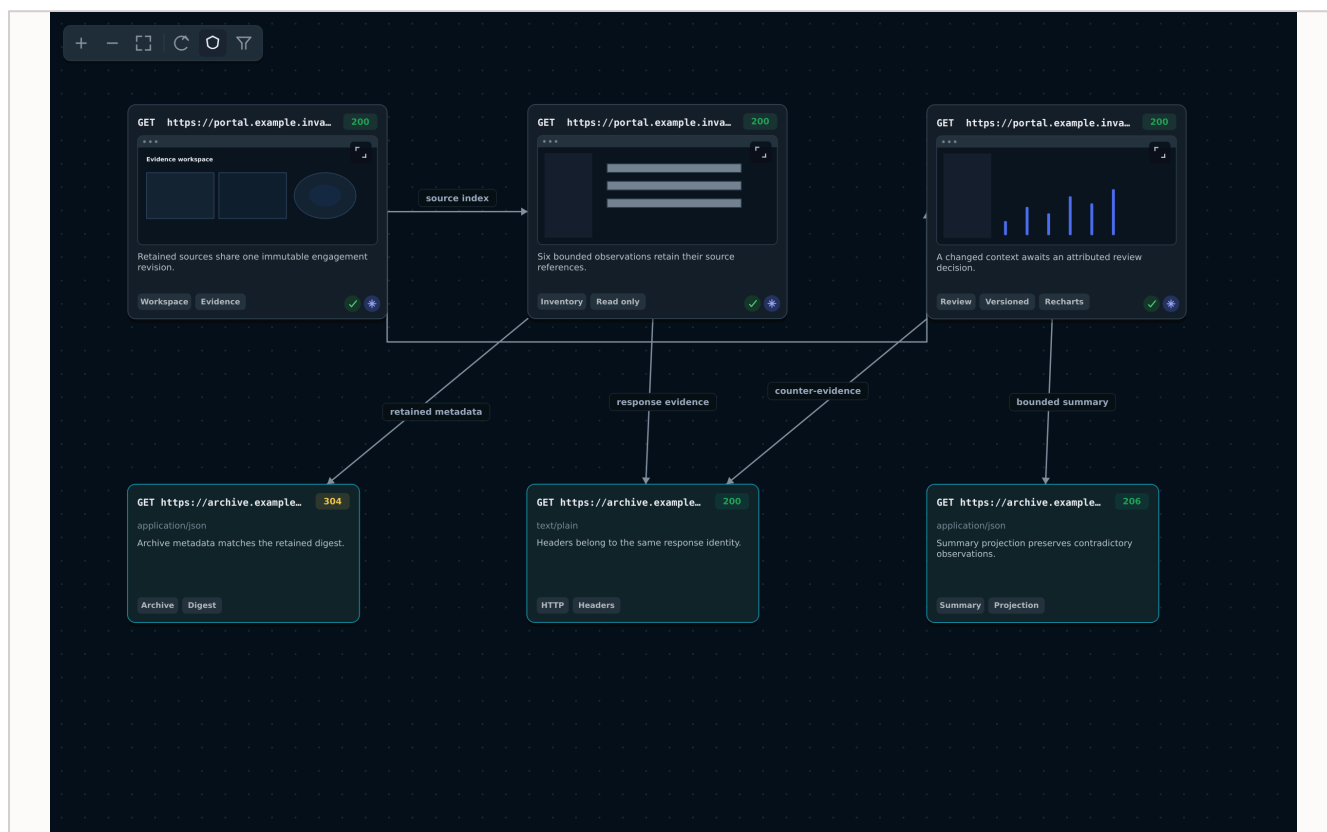
Reference 05

authoritative engagement architecture

EVIDENCE ATTRIBUTION

From tool output to reusable evidence

Integrated tools contribute observations with their source, context and interpretation retained. Source references make an accepted result useful to the next stage, the next model segment and the operator inspecting the decision.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

Evidence categories support correlation without flattening distinct candidates into one summary. This is the connection between broad tool coverage and coherent automated reasoning.

Implemented

Reference 06

canonical specialist contracts

CONDITIONAL STAGE COVERAGE

The Core graph contains 88 conditional stages

Discovery, service and identity analysis, web investigation, artifact analysis, session work and reporting are represented in 88 automatic stage declarations. A separate set of 13 C2 declarations adds common and platform-specific session families.

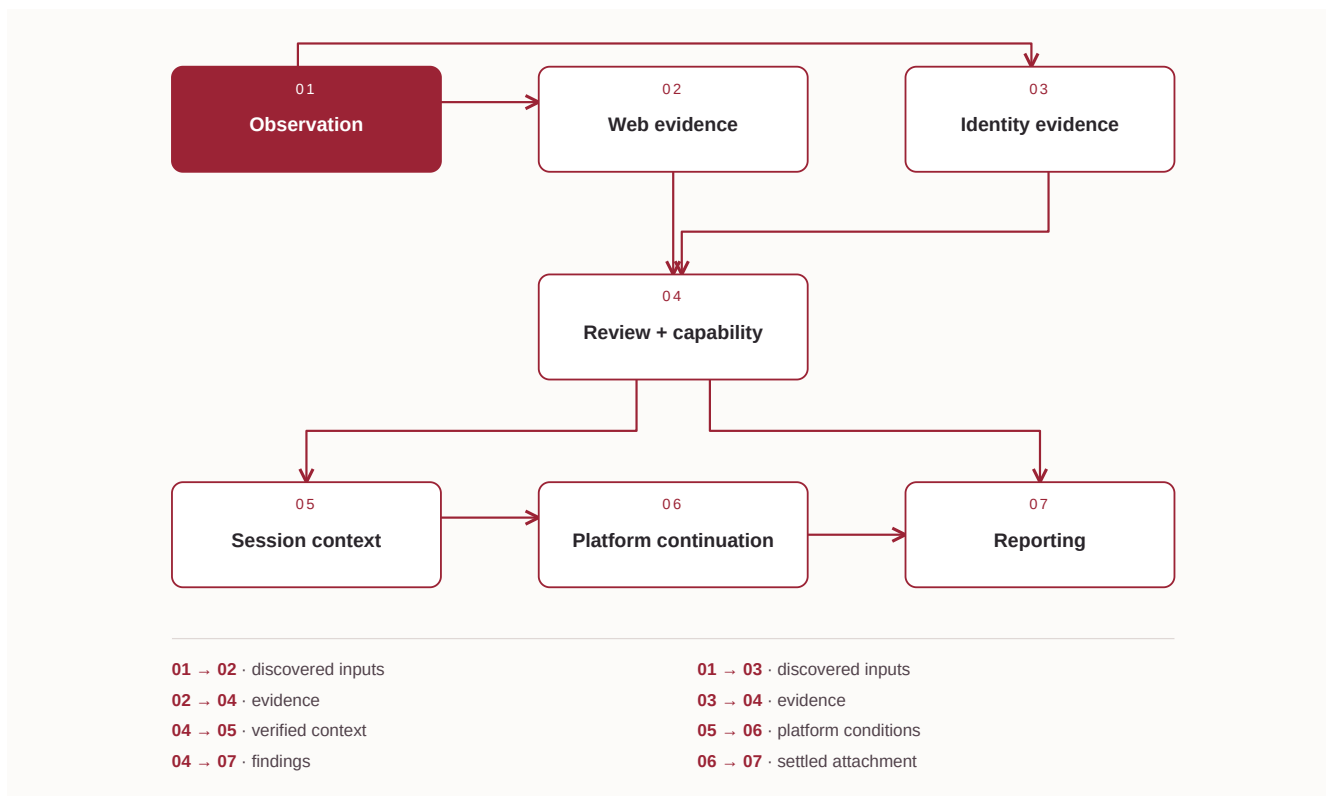


Diagram of the labeled component relationships.

Prerequisites, evidence, platform selection and operator policy select the actual path. The stage atlas provides every automatic declaration with its source label and input/output role; 88 does not describe a mandatory linear execution sequence.

Implemented

Reference 07

`semantic_registry.go + chain_runtime.go`

DEPENDENCY AND EVIDENCE RELATIONSHIPS

The scheduler separates dependencies from correlations

The Execution Graph distinguishes scheduling dependencies from evidence correlations. Phase barriers, declared branch dependencies and scoped eligibility control work, while context and evidence links enrich the operator's view.

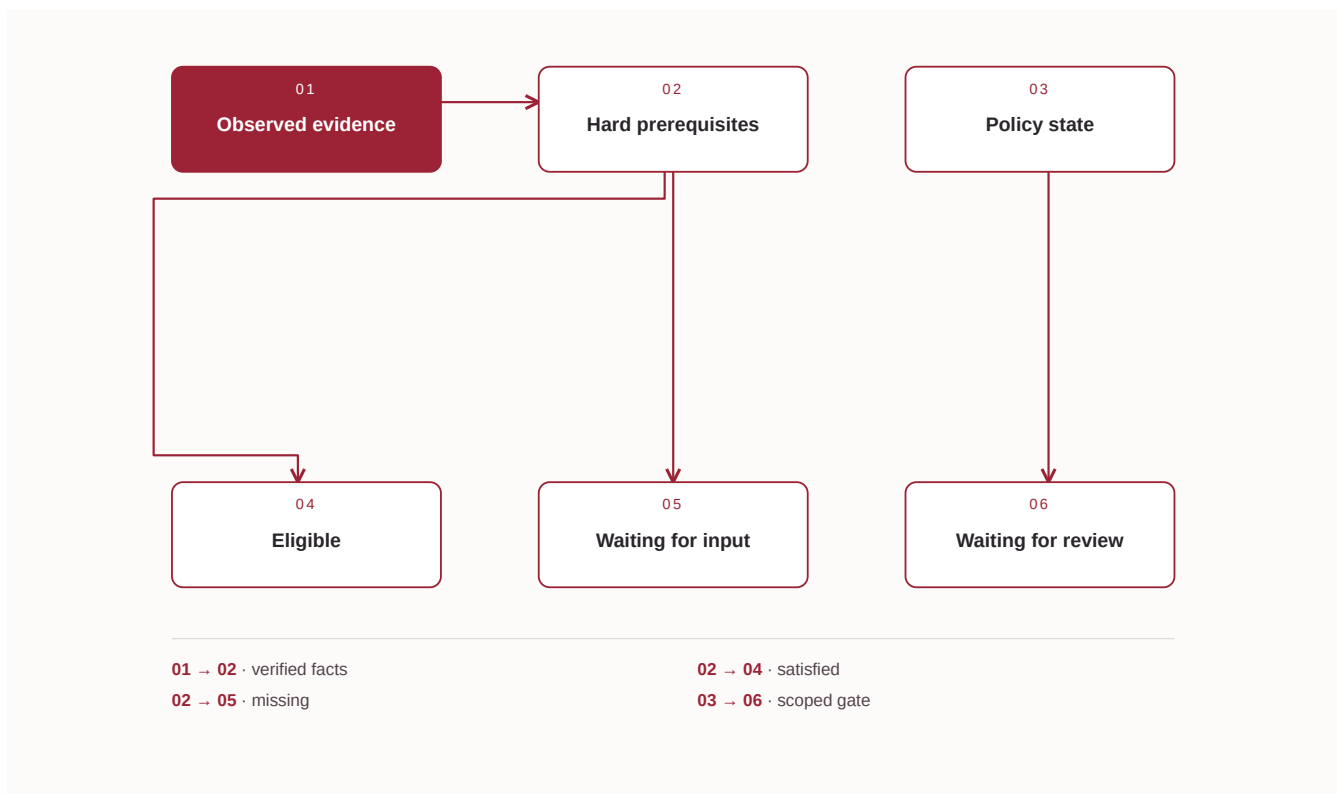


Diagram of the labeled component relationships.

The scheduler evaluates a bounded stage view. Dynamic endpoint and gate nodes remain useful to the operator without multiplying the work required to decide which stage is ready.

Implemented

Reference 08
execution graph lifecycle

BRANCH-LOCAL PROGRESS

Independent branches progress within their own scope

The scheduler separates branch-local gates from a global pause. A web review, identity question or pending prerequisite can wait in its own scope while other eligible work advances.

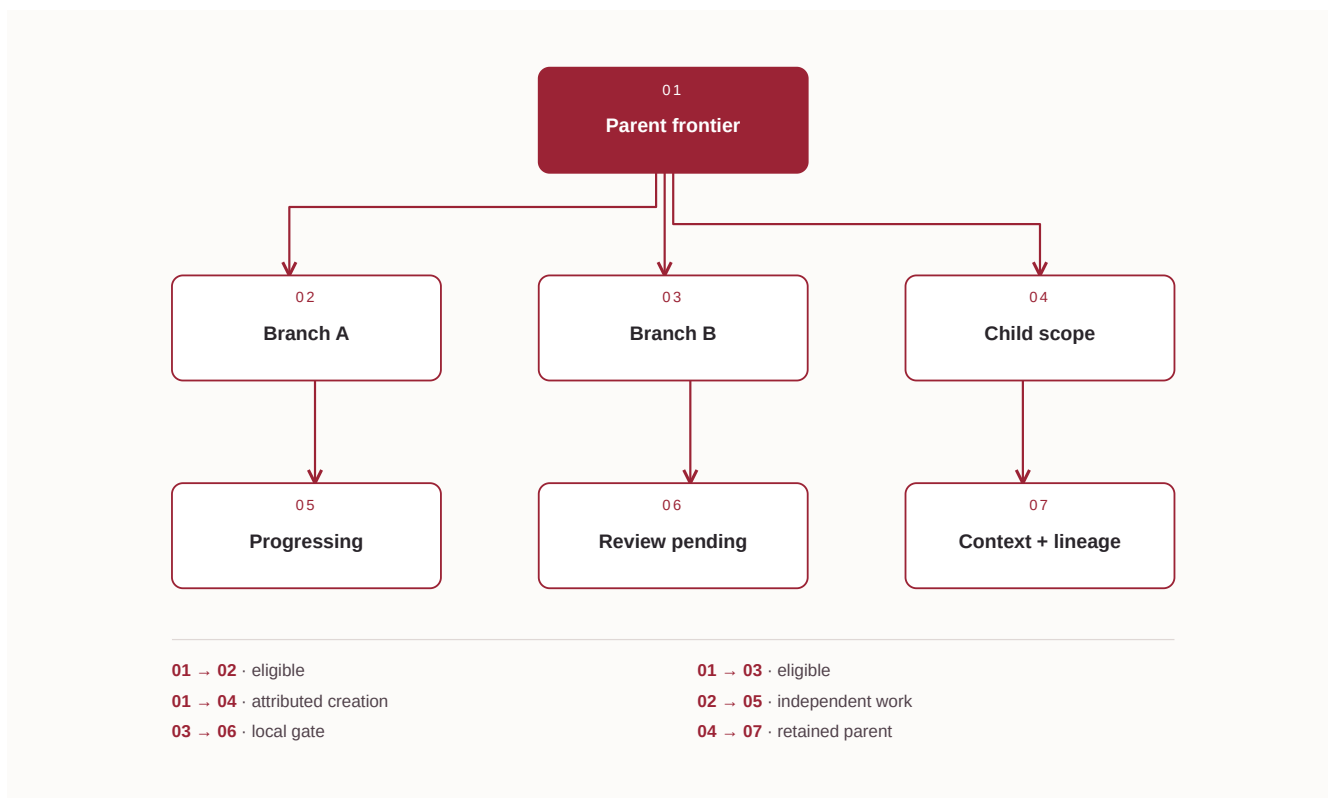


Diagram of the labeled component relationships.

Loops and dormant branches respond to new evidence. The frontier records which input can change eligibility, giving long investigations a clear continuation point.

Implemented

Reference 09
execution graph lifecycle

BOUNDED MODEL CALLS

Specialist calls operate within defined task boundaries

Bounded stateless AI calls receive a defined purpose and a structured evidence packet. Core includes a bank of 100 fine-tuned specialist adapters for these focused calls and code-review roles. Deterministic code checks the response, attributes the model and retains accepted results.

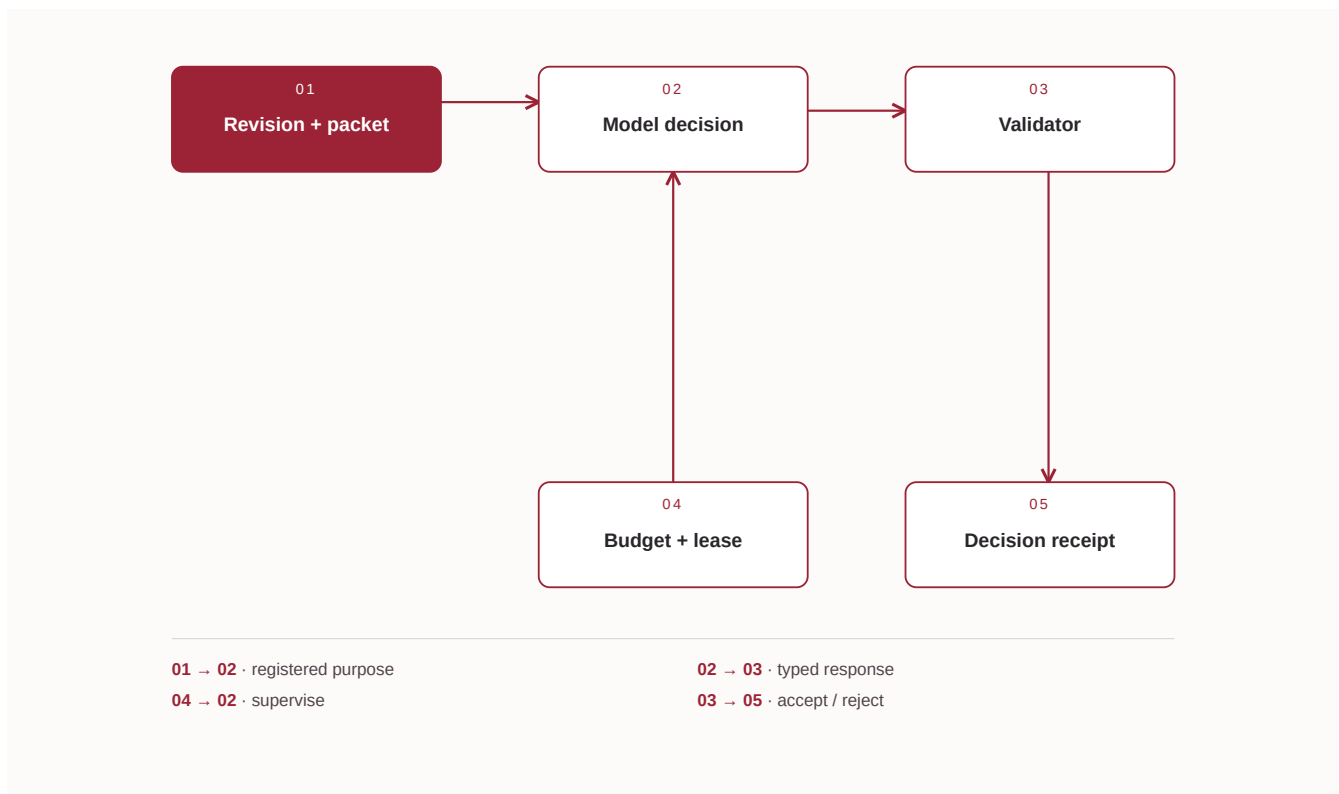


Diagram of the labeled component relationships.

Token accounting, capacity controls and response validation surround each call. Specialist selection follows the task boundary while the workspace retains its evidence and review history.

Core model entitlement

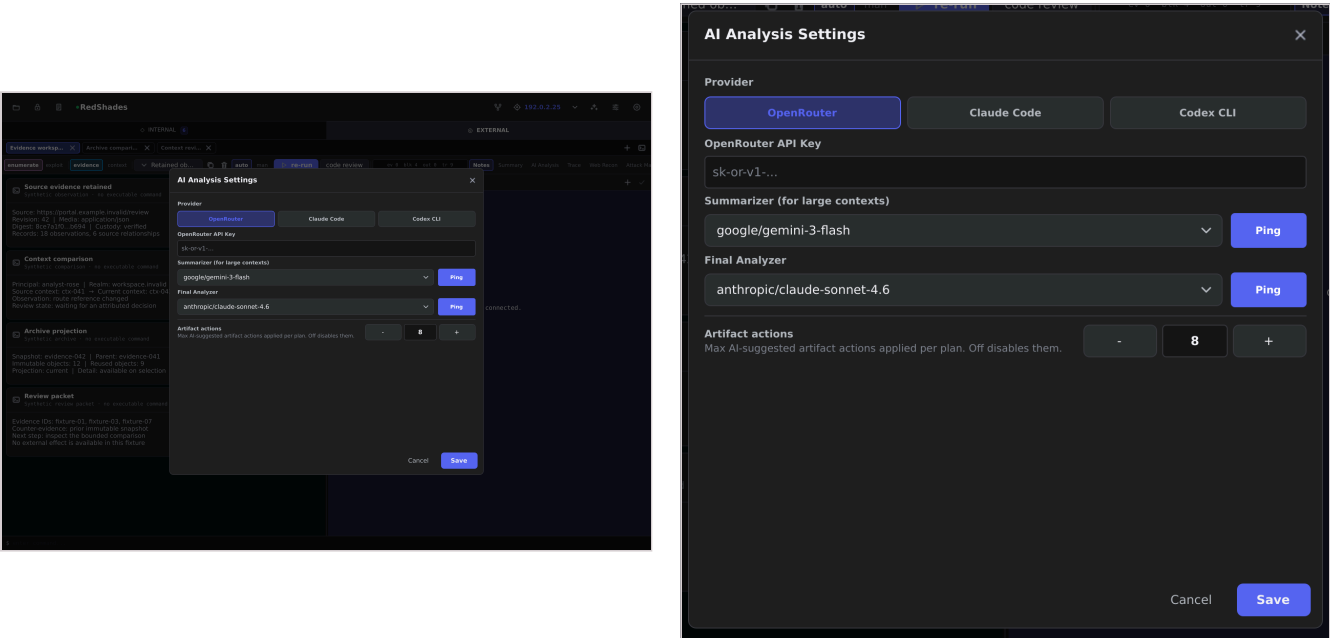
Reference 10

commercial model specifications

EXECUTIVE SUPERVISION

The Executive supervises progress across stages

The Executive observes progress across stages, prioritizes attention and routes capability gaps or engineering incidents. Its persistent supervisory context complements the short, focused calls used inside individual boundaries.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data. Overview and detail crop from identical source pixels.

Executive, ACP and AEP workspaces preserve artifacts and checkpoints for longer tasks. Model capacity can be allocated by responsibility, while the deterministic controller retains authority over the workflow.

Implemented

Reference 11
executive teacher babysitter plan

PERSISTENT MODEL SESSIONS

Persistent sessions for more complex work

Advanced includes a fine-tuned 30B3A Flash model for persistent higher-complexity sessions. Executive, ACP and AEP workspaces retain accepted artifacts, prior review and progress records between segments, supporting work that benefits from continuity.

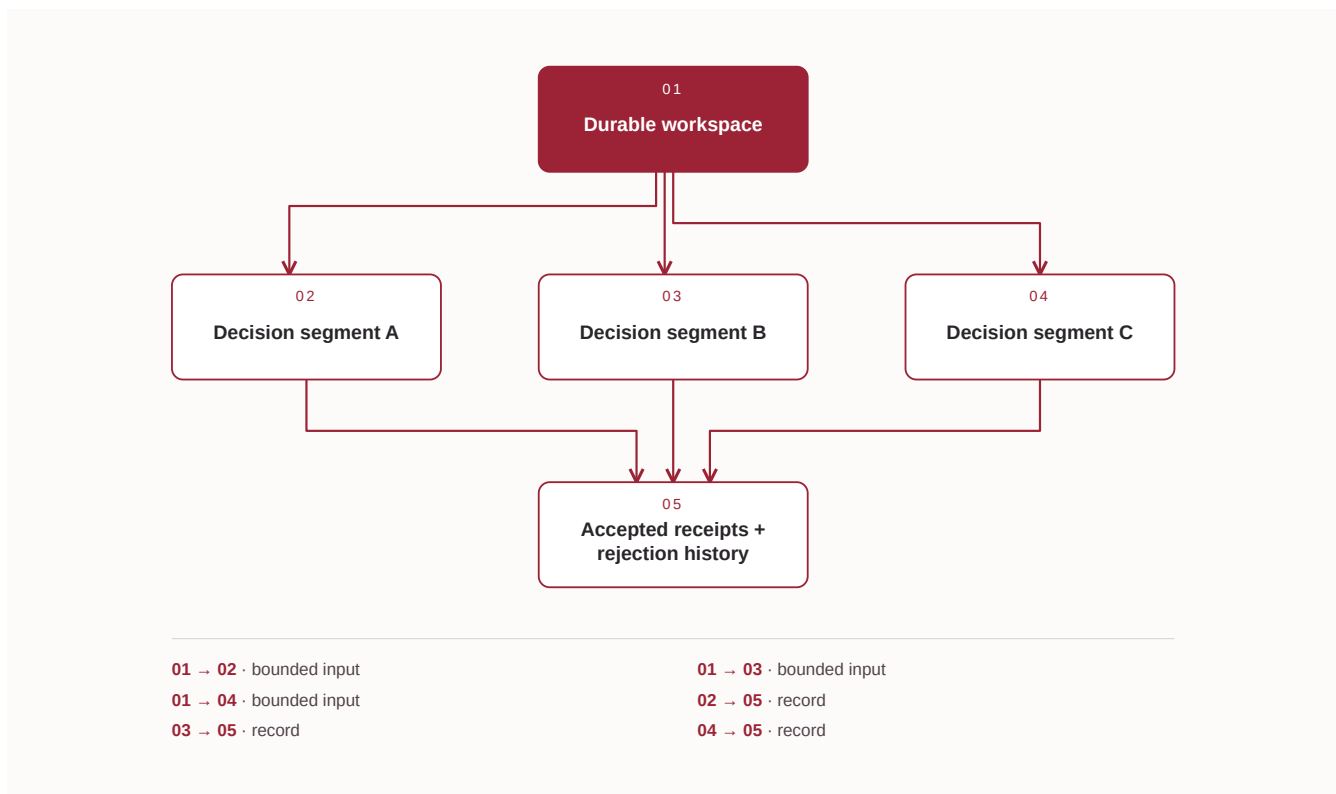


Diagram of the labeled component relationships.

The model complements bounded specialist calls within the same orchestration architecture. Model and adapter weights are provided under NDA; active subscriptions receive weights and stage updates at least every six months.

Advanced model entitlement

Reference 12

commercial model specifications + canonical specialist

contracts

CAPABILITY DEVELOPMENT

ACP builds evidence-bound capabilities

The Adaptive Capability Pipeline develops custom capability candidates from observed evidence and source mechanics. A persistent workspace owns binding, SDK selection or extension, adapter generation, review and replica validation.

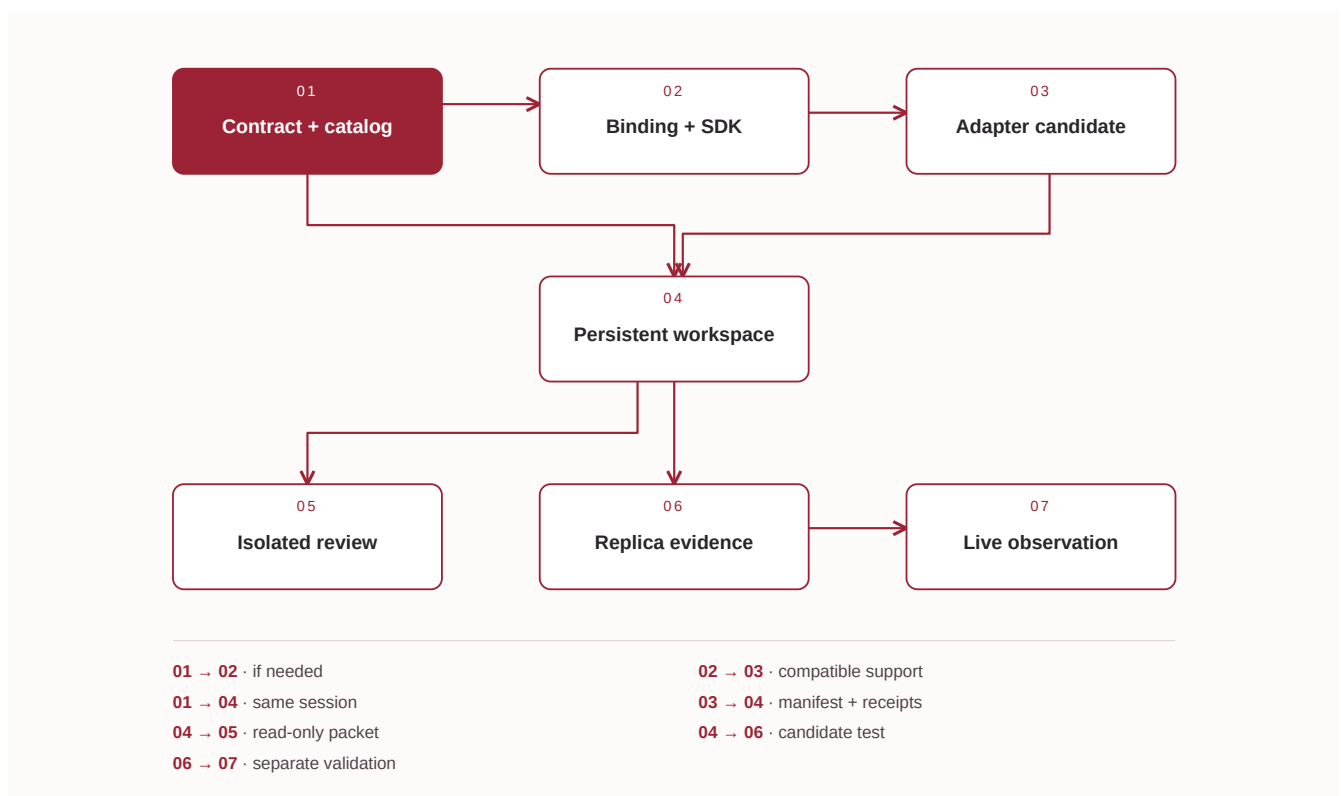


Diagram of the labeled component relationships.

Compatible catalog artifacts and SDKs can skip generation segments. New work retains its contract and review history, connecting flexibility with reusable, reviewed artifacts.

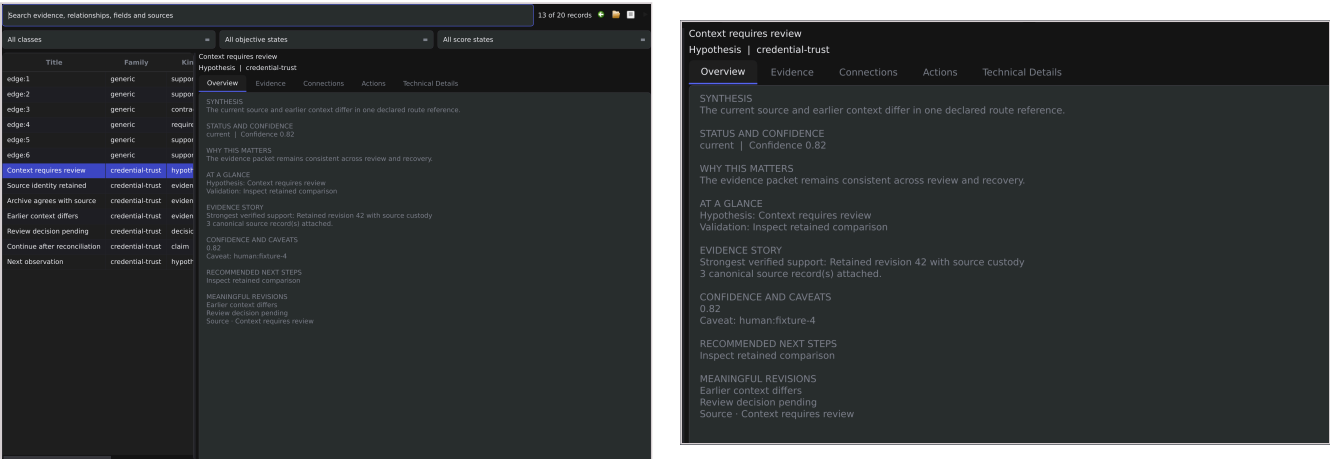
Implemented

Reference 13
acp vertical session

CANDIDATE PUBLICATION

From an advisory to a reviewed candidate

ACP connects source findings and advisories with typed capability contracts, SDK extensions and reviewed adapters. Where a ready-made proof of concept is absent, the workbench can retain a candidate for controlled evaluation.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data. Overview and detail crop from identical source pixels.

Candidate source, metadata and tests are published as a digest-bound set. Admission and isolated review establish the boundary between generated material and an approved artifact.

Implemented

- Reference 14
- acp vertical session

INDEPENDENT CAPABILITY REVIEW

Review is built into capability production

An isolated reviewer evaluates the candidate against its contract and provenance. Counterfactuals test whether changed names, ordering and evidence still support the same capability, helping turn a one-off result into a reusable component.

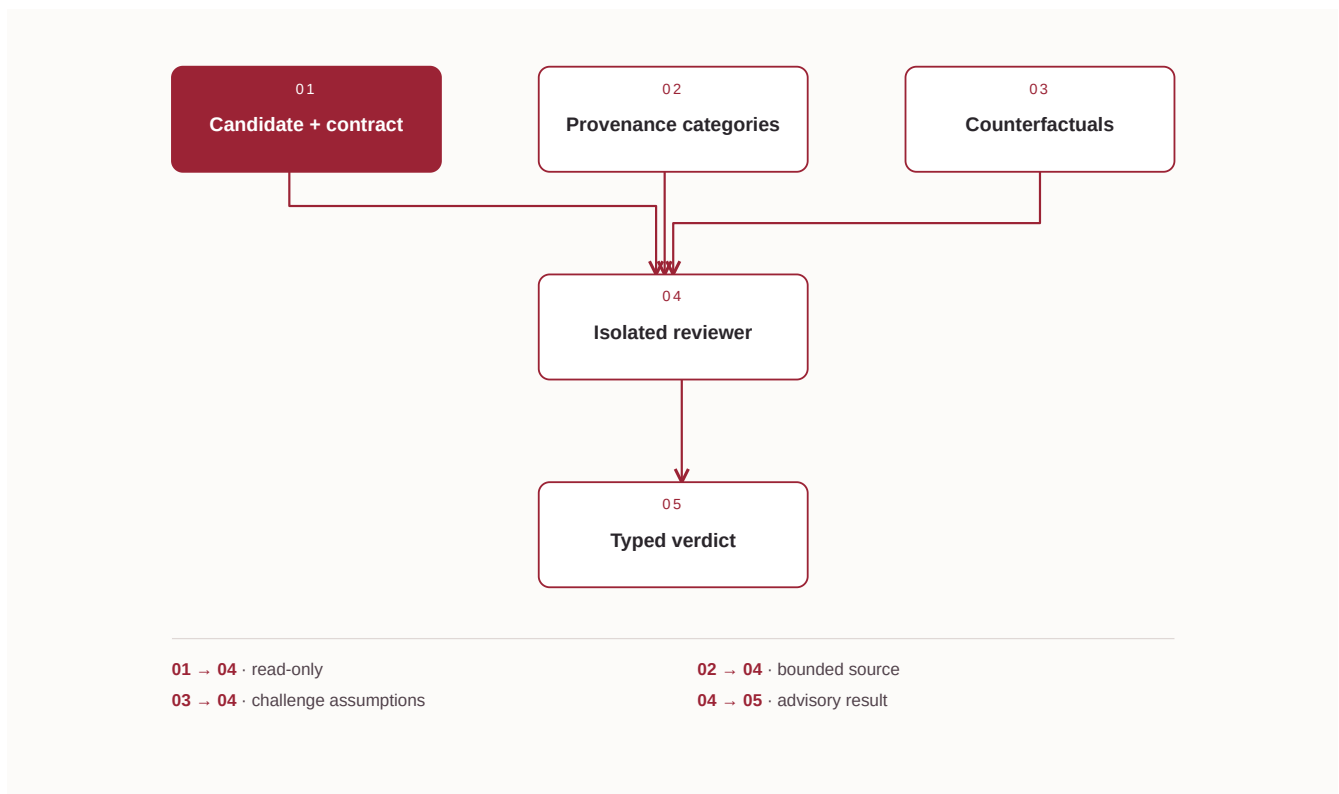


Diagram of the labeled component relationships.

The reviewer uses a read-only packet and returns a typed verdict to admission. Generation, review and execution authority remain separately owned.

Implemented

Reference 15
acp vertical session

RUNTIME SOFTWARE REPAIR

AEP repairs software while the run is in flight

The Autonomic Engineering Plane owns diagnosis, a persistent repair session, scoped source changes, tests, build admission, hot replacement and verified hand-back. It addresses defects in the platform machinery while ACP addresses missing target interactions.

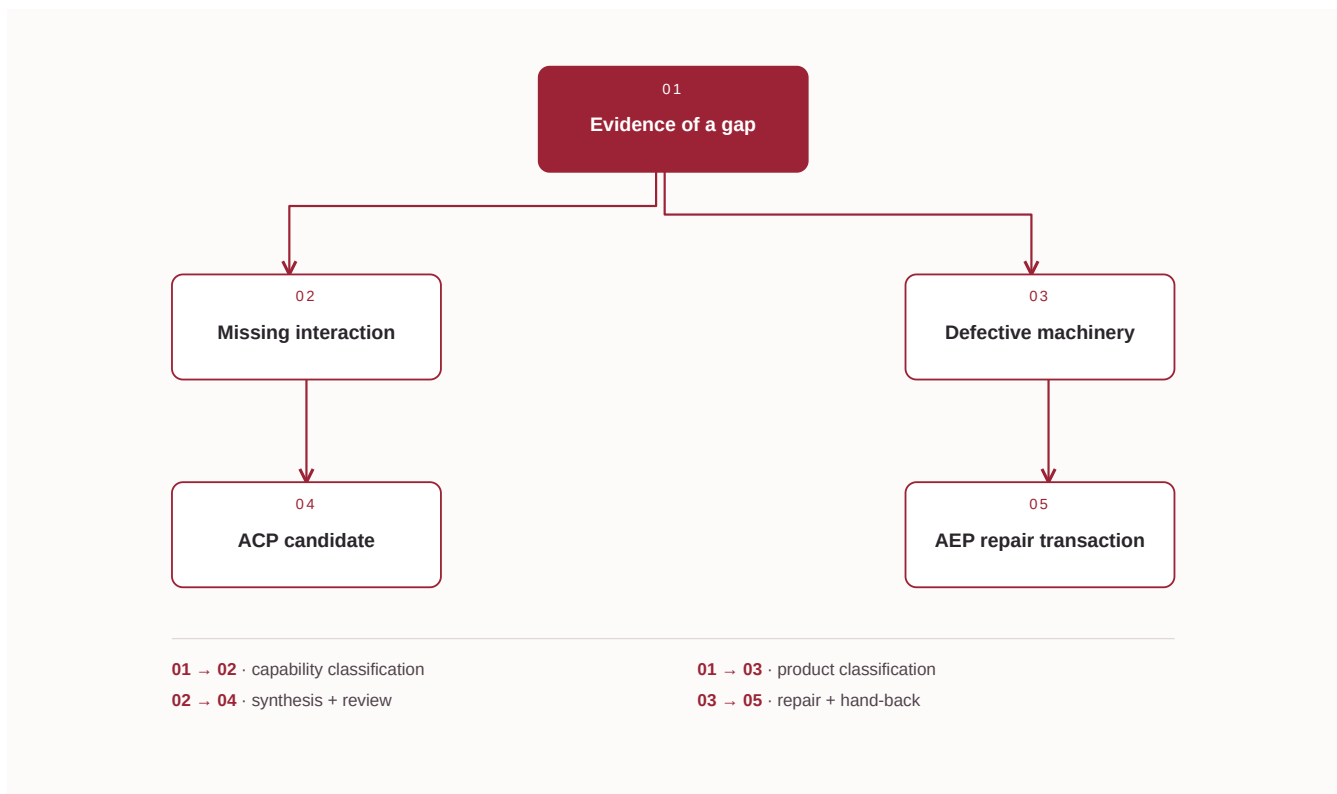


Diagram of the labeled component relationships.

The full repair transaction preserves the run's context through a provider replacement. Its acceptance is defined by validated hand-back, connecting software engineering work to continued execution.

Implemented

Reference 16

executive teacher babysitter plan

SPECIALIST MODEL ROLES

100 fine-tuned specialist adapters

Core includes a bank of 100 fine-tuned specialist adapters supporting bounded stateless calls and code-review roles. Focused model work sits within the platform's evidence, workflow and response-validation boundaries.

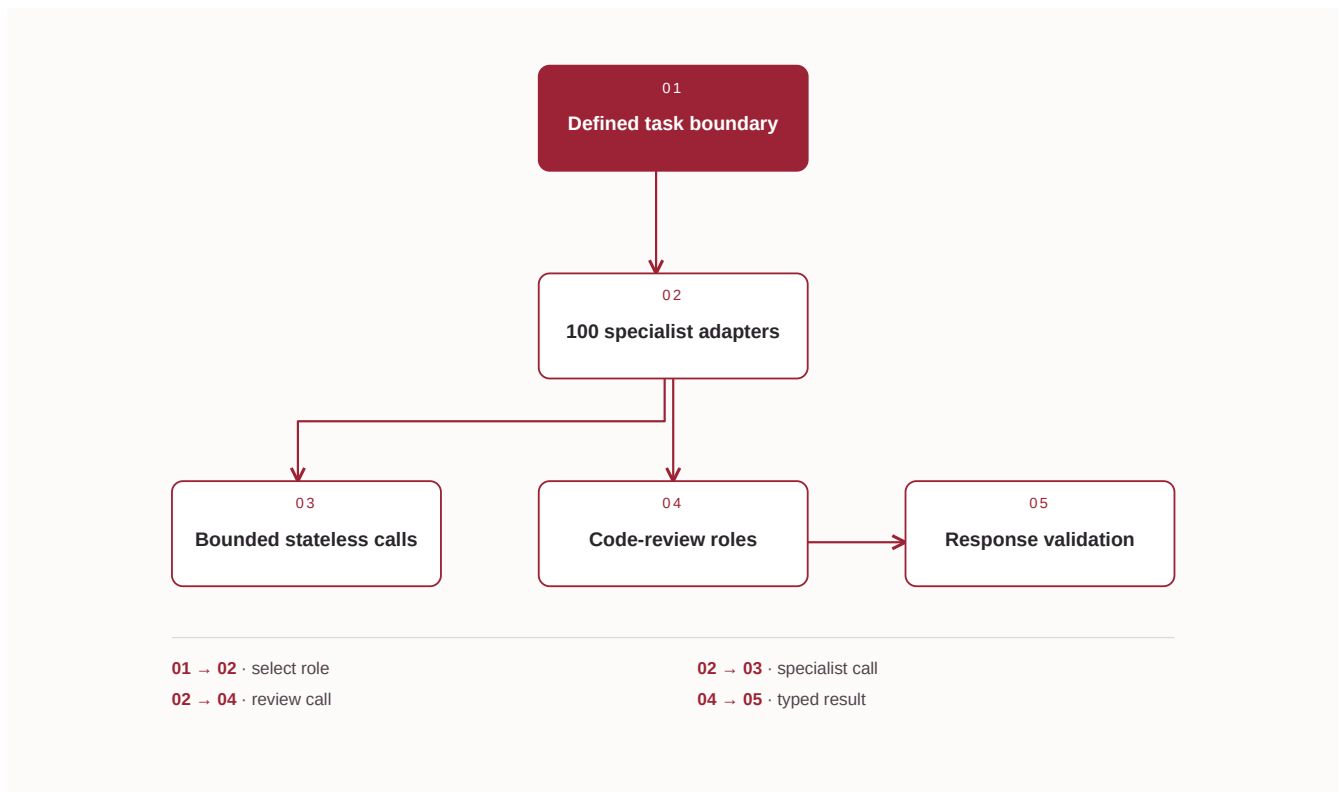


Diagram of the labeled component relationships.

Ultimate adds a proprietary multi-encoder neural representation suite for defensive and forensic workloads. These task-specific embedding and representation networks have multiple workload variants. They are separate from the language models used for stateless calls and persistent sessions.

Commercial model entitlements

Reference 17
commercial model specifications

PROVIDER REPLACEMENT

Provider replacement preserves engagement state

The Runtime Composition Kernel publishes versioned providers and controls leases, rollback and cleanup. New work can adopt a replacement while existing work retains its implementation and engagement records.

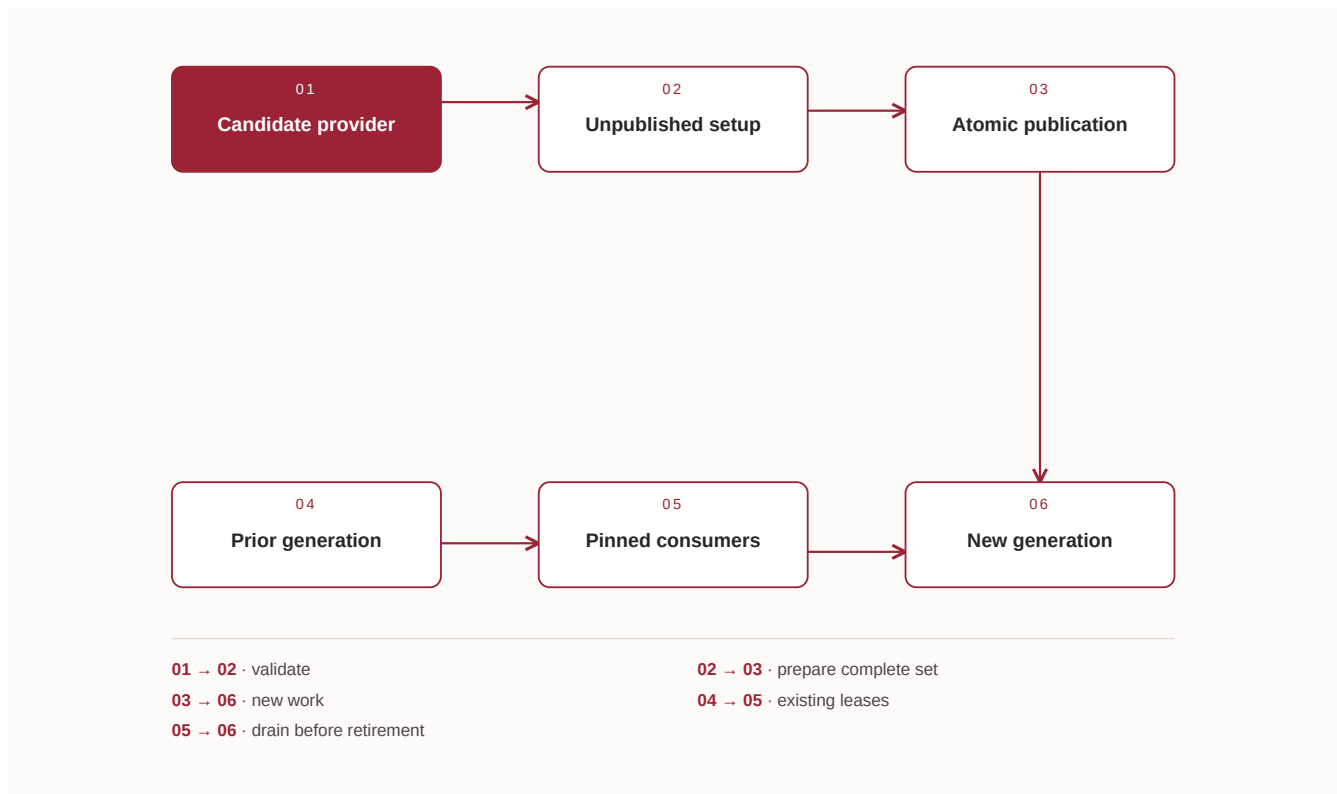


Diagram of the labeled component relationships.

This is the runtime foundation beneath AEP hand-back. State remains outside replaceable behavior, giving repair a defined boundary rather than requiring a whole-workspace restart.

Implemented

Reference 18
runtime composition kernel

WORKER RESULT ATTRIBUTION

Workers return results under a lease

Isolated stage workers receive immutable execution envelopes. Results are accepted only against the current artifact, input and generation lease. That prevents an old worker from publishing a plausible result after its authority has expired.

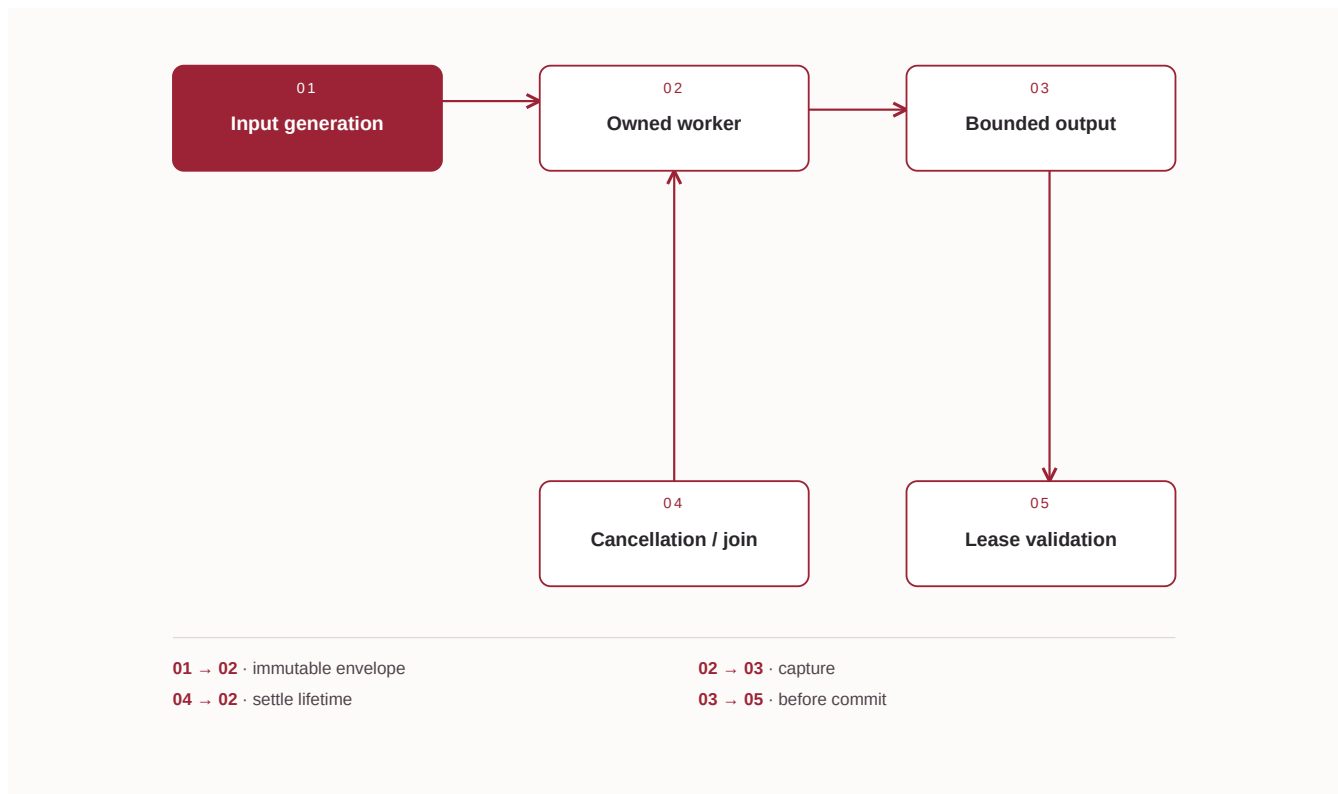


Diagram of the labeled component relationships.

In-process handlers and isolated workers have different replacement boundaries. The catalog describes the common principle without implying that every component can be replaced at any arbitrary instruction.

Implemented

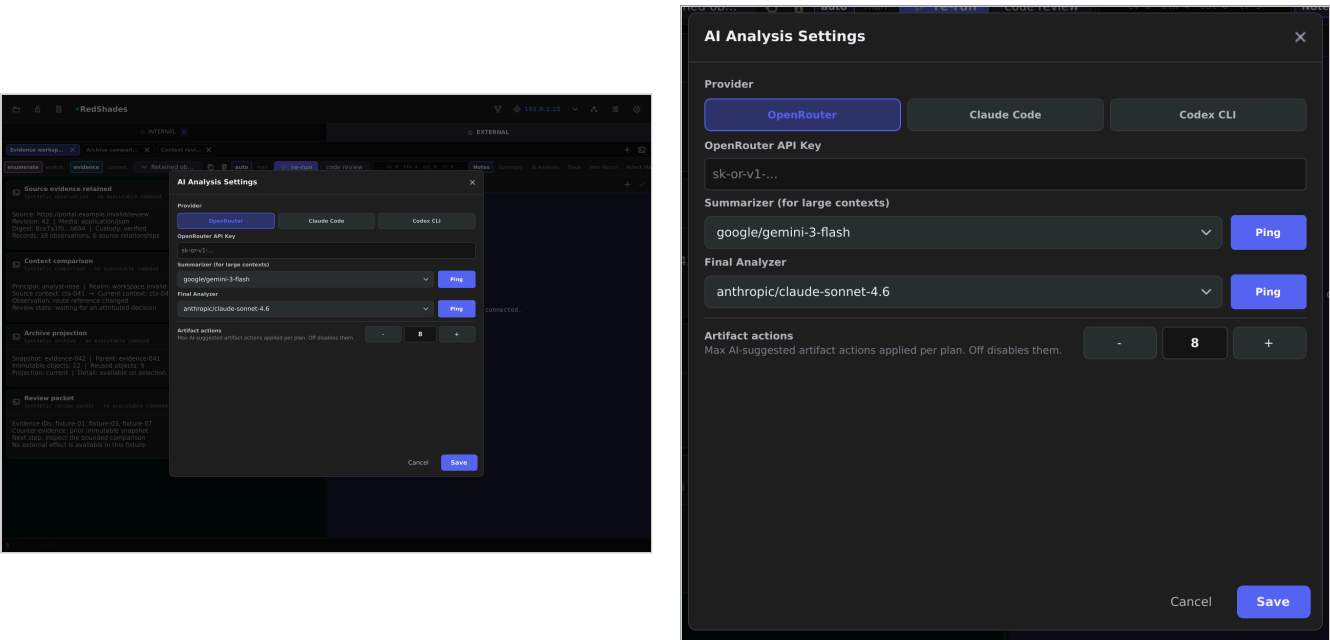
Reference 19

hot swappable stage workers

RESOURCE OWNERSHIP

Context and resource budgets stay visible

Custom context management and token accounting track the cost of model work. Resource supervision retains ownership of worker processes, bounded output and cleanup, supporting long sessions without losing sight of their compute footprint.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data. Overview and detail crop from identical source pixels.

Token and resource diagnostics make unusual consumption inspectable. Named owners and retained cleanup handles let the platform distinguish active work from stale work.

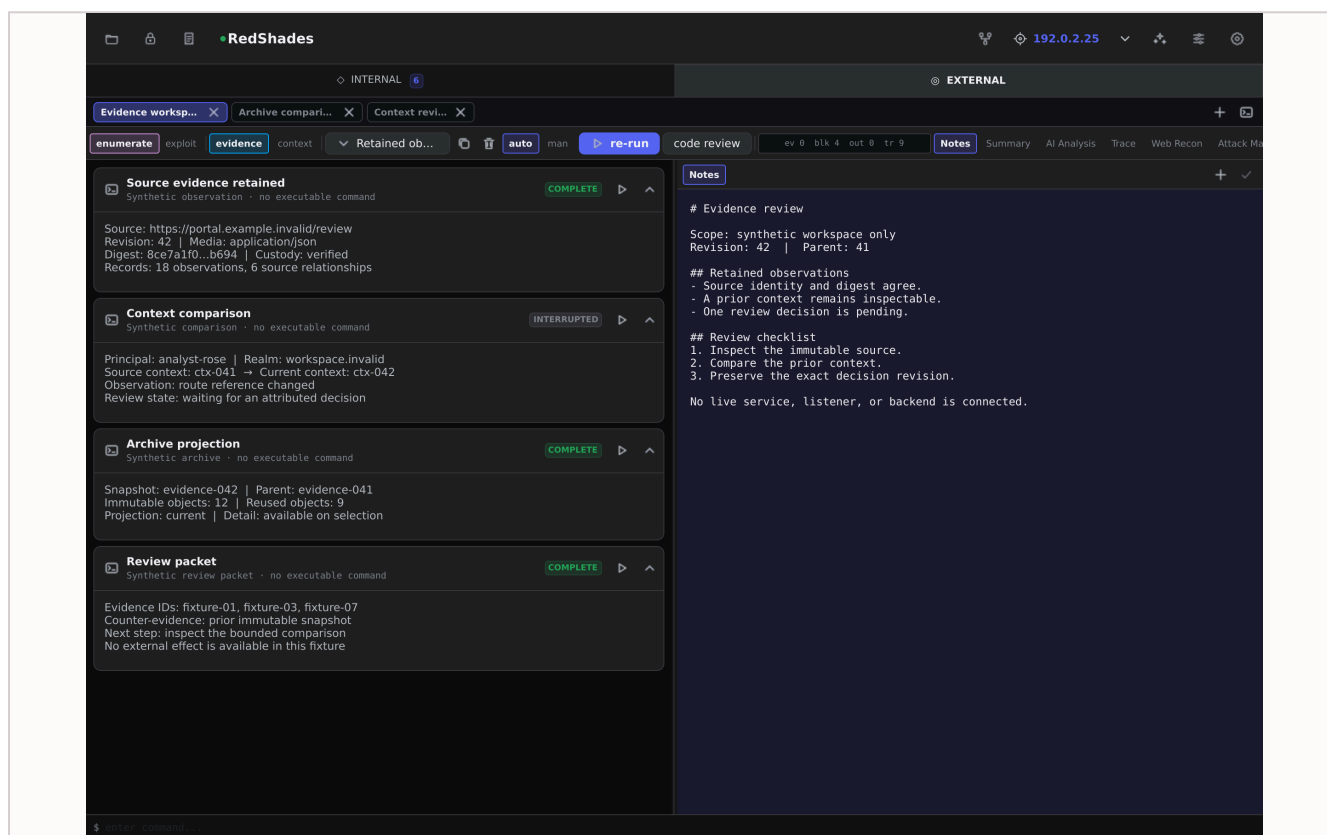
Implemented supervision boundaries

Reference 20
service process custody

WORKSPACE INTEGRATION

Qt, watchdog and tools in one workspace

The native Qt operator places dense graphs and investigation panels inside a managed desktop workspace. The watchdog hosts retained service and tool surfaces, connecting Gitea, BloodHound, build workflows, RDP and development harnesses with the operator environment.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

Lightweight QML chrome surrounds native widgets. Embedded tool windows and custom analysis views can sit alongside each other, preserving tool fidelity while keeping the investigation visible.

Implemented

Reference 21
subsystem glossary

THIRD-PARTY EXECUTION

Third-party tools return evidence through isolated execution

Tool descriptors connect network, web, directory, browser and source-inspection families to the platform. Admitted third-party proof-of-concept work can use isolated container environments, with outputs returned as attributed evidence.

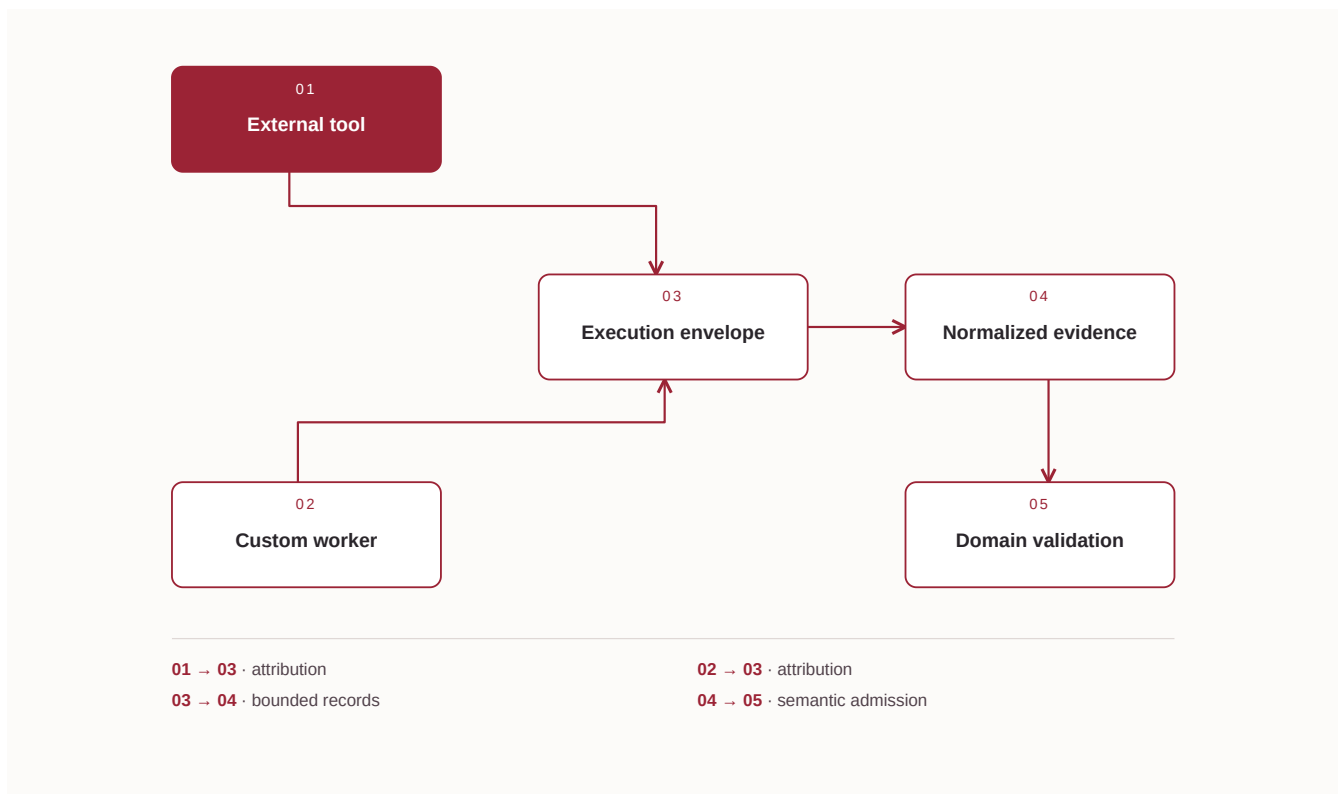


Diagram of the labeled component relationships.

Integration combines execution policy, process ownership and evidence normalization. Tool availability, candidate review and permission remain distinct inputs to the workflow.

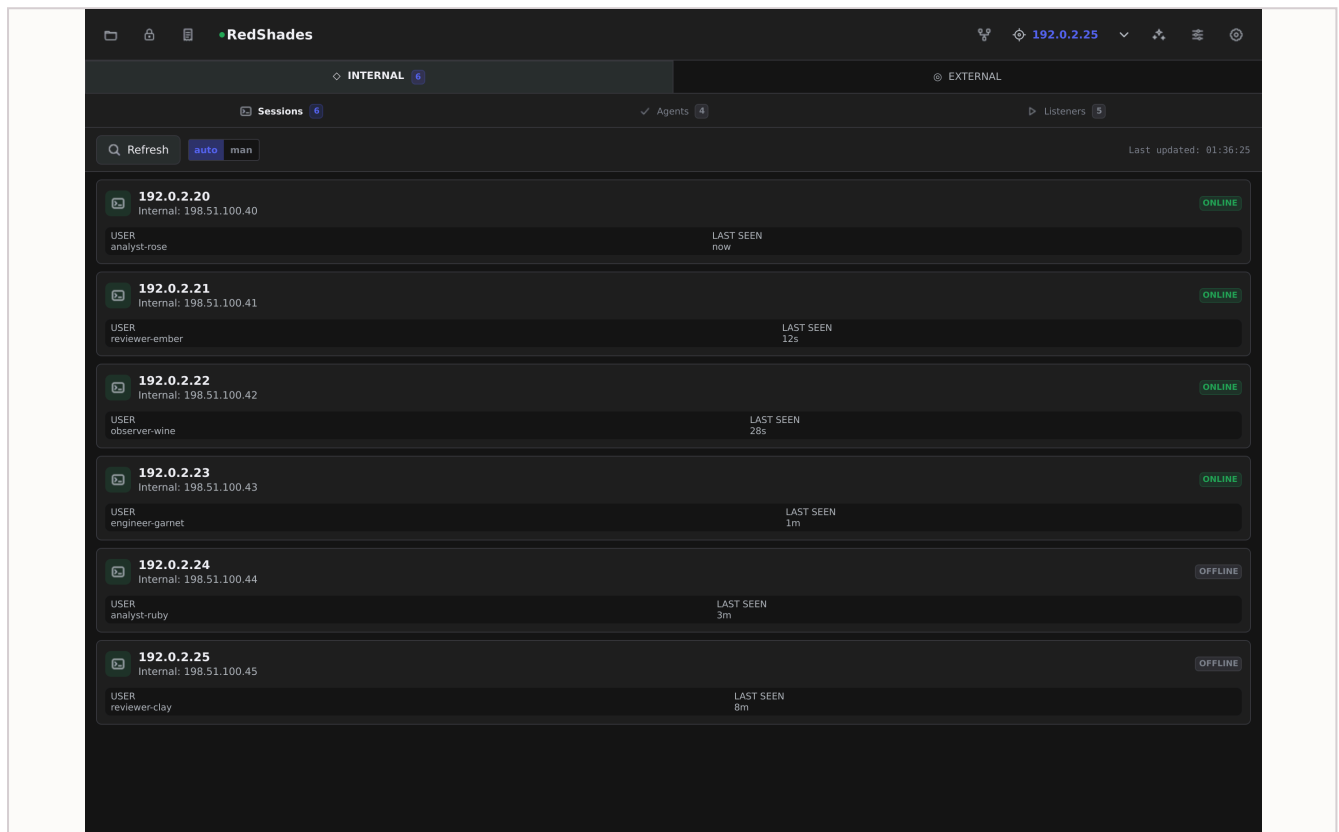
Implemented

Reference 22
specialist plugin registry

ARTIFACT IDENTITY

VM-based transformation and artifact engineering

The artifact and implant toolchain includes transformation and obfuscation work alongside versioned builds. VM-oriented transformation is part of the project's engineering breadth, with artifact identity and provenance carried into the wider workspace.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

The publication describes these systems at component level. Build records identify the material and its role; native artifact surfaces keep the resulting inventory accessible to the operator.

Implemented

Reference 23

deployment and process ownership

SESSION CONTINUATION

Session work extends the investigation

C2 stage families add platform-aware continuation once a session is registered. Context, artifacts and attached work remain connected to the parent engagement, extending the same orchestration model beyond initial observation.

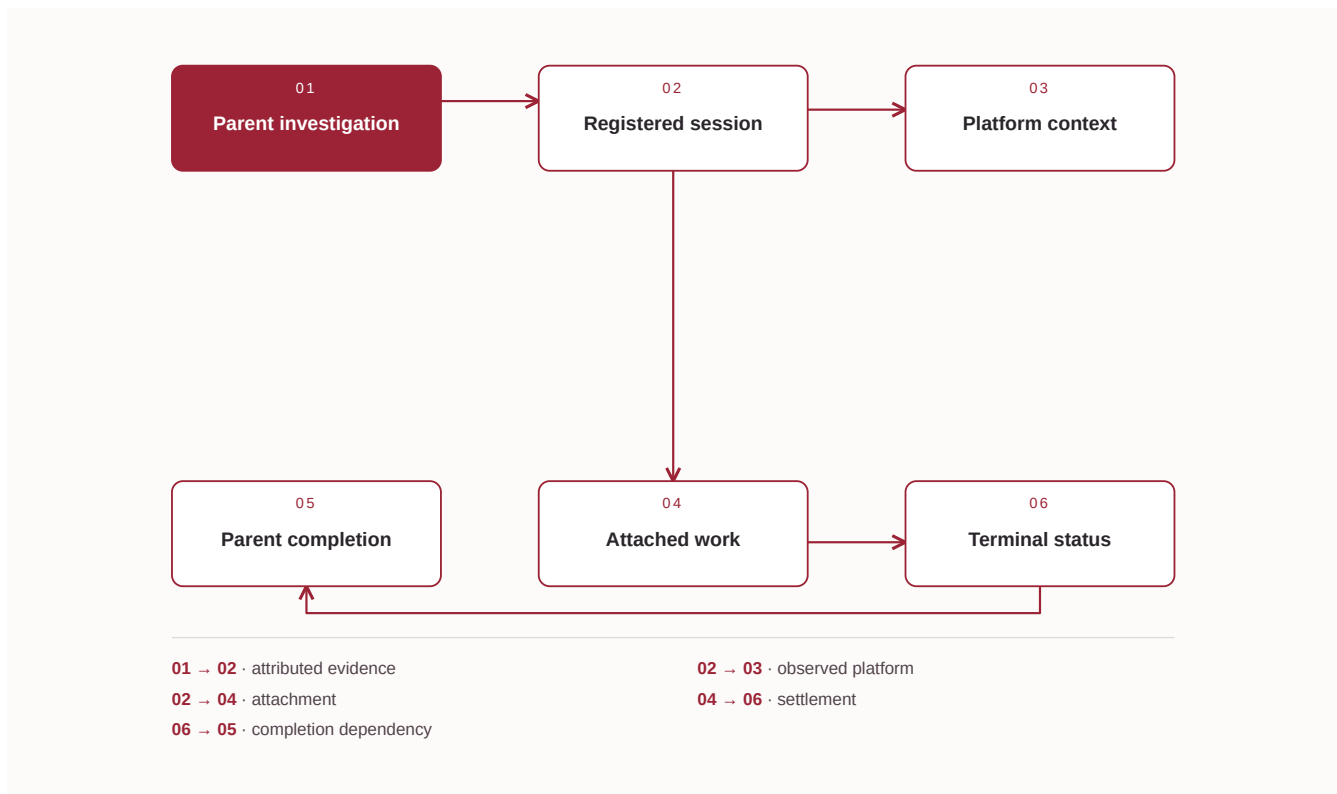


Diagram of the labeled component relationships.

Common and Windows/Linux declarations preserve their own prerequisites. The automatic stage atlas and the C2 declaration inventory describe separate parts of the runtime graph.

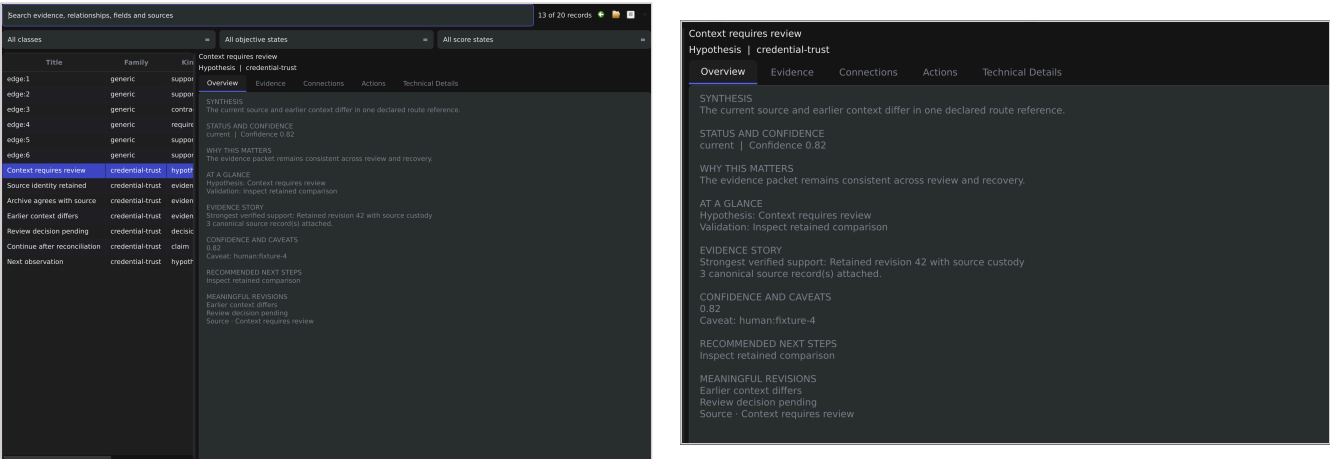
Implemented

Reference 24
authoritative engagement architecture

EMBEDDED SPECIALIST TOOLS AND CUSTOM ANALYSIS

Reverse engineering inside the workflow

Binary and source analysis connect to the investigation workbench and embedded specialist tools. Custom reverse-engineering workflows can sit beside native evidence views and retained external tool surfaces.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data. Overview and detail crop from identical source pixels.

The integrated workspace brings analysis, artifact records and engineering context together. Tool-specific interfaces remain available while their results feed the shared investigation record.

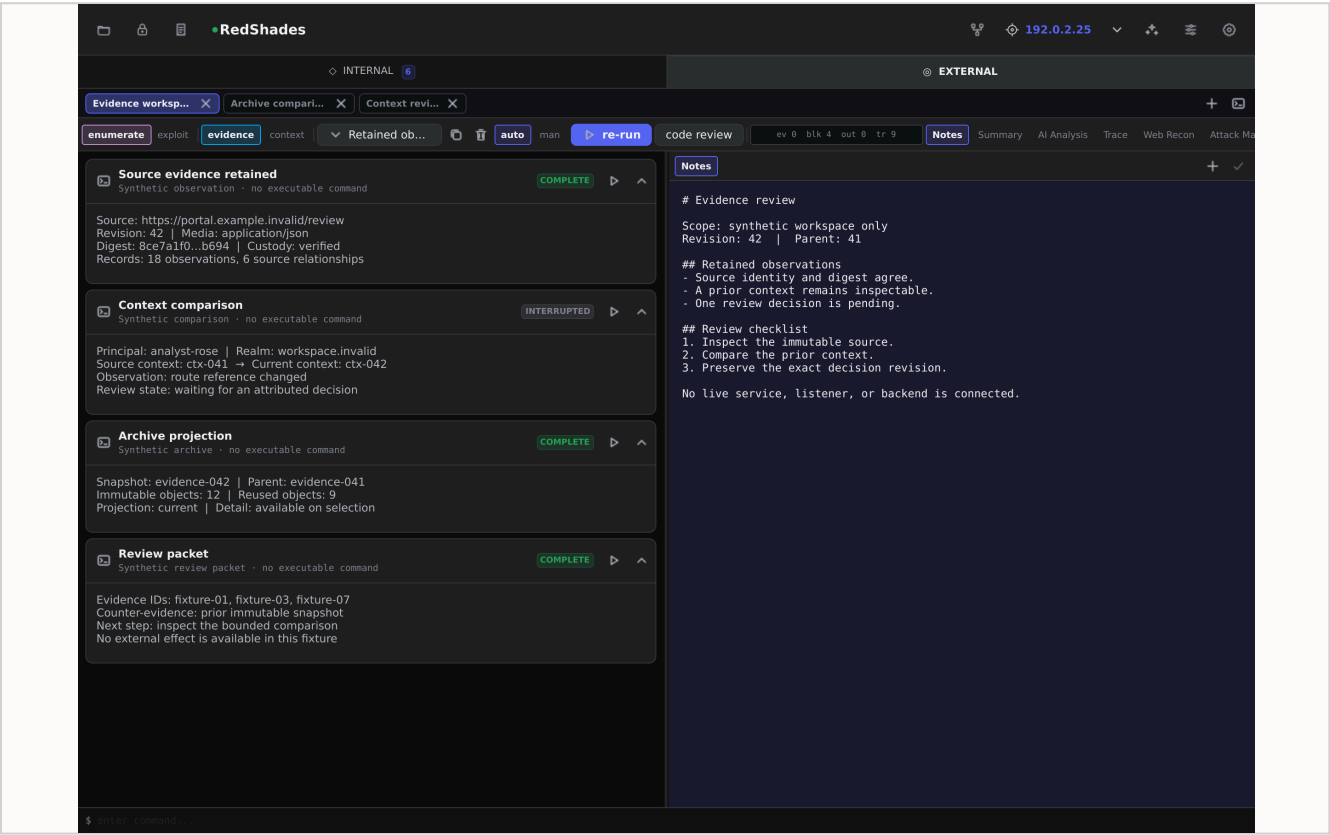
Implemented

- Reference 25
- execution graph lifecycle

OPERATOR WORKSPACE

A working desktop for the whole investigation

Network Graph, WebRecon, Investigation Graph, session records and model controls share a native Qt operator. Embedded tools extend the workspace rather than forcing every workflow into a generic dashboard.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

The retained screenshots show real Qt widgets with fictional disconnected data. Their arrangement demonstrates how overview, selected evidence and specialist work can remain visible together.

Native interface

Reference 26
subsystem glossary

NETWORK GRAPH

Network Graph displays hosts and their relationships

The Network Graph presents observed hosts and relationships spatially. Its value is orientation: which context is selected and how that context relates to the wider investigation.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

Selecting a host or relationship connects the topology view to the investigation context shown in the workspace.

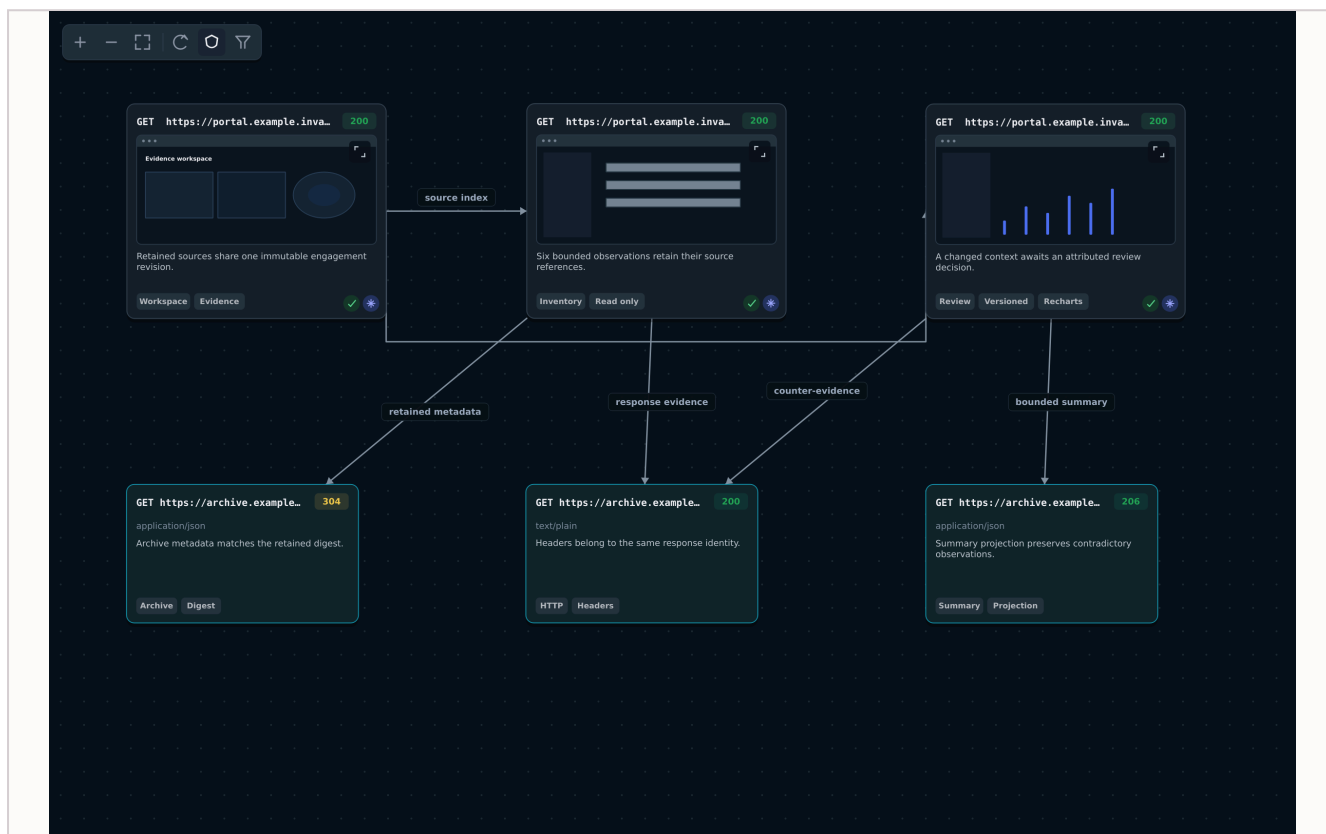
Native interface

Reference 27
subsystem glossary

WEBRECON OVERVIEW

WebRecon links observations to their source material

WebRecon brings web observations, source detail and endpoint review into a native workspace. A useful arrangement lets the operator retain overview while examining the evidence behind one selected item.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

The overview and source material remain visible within the same workspace, preserving the context behind an observation.

Native interface

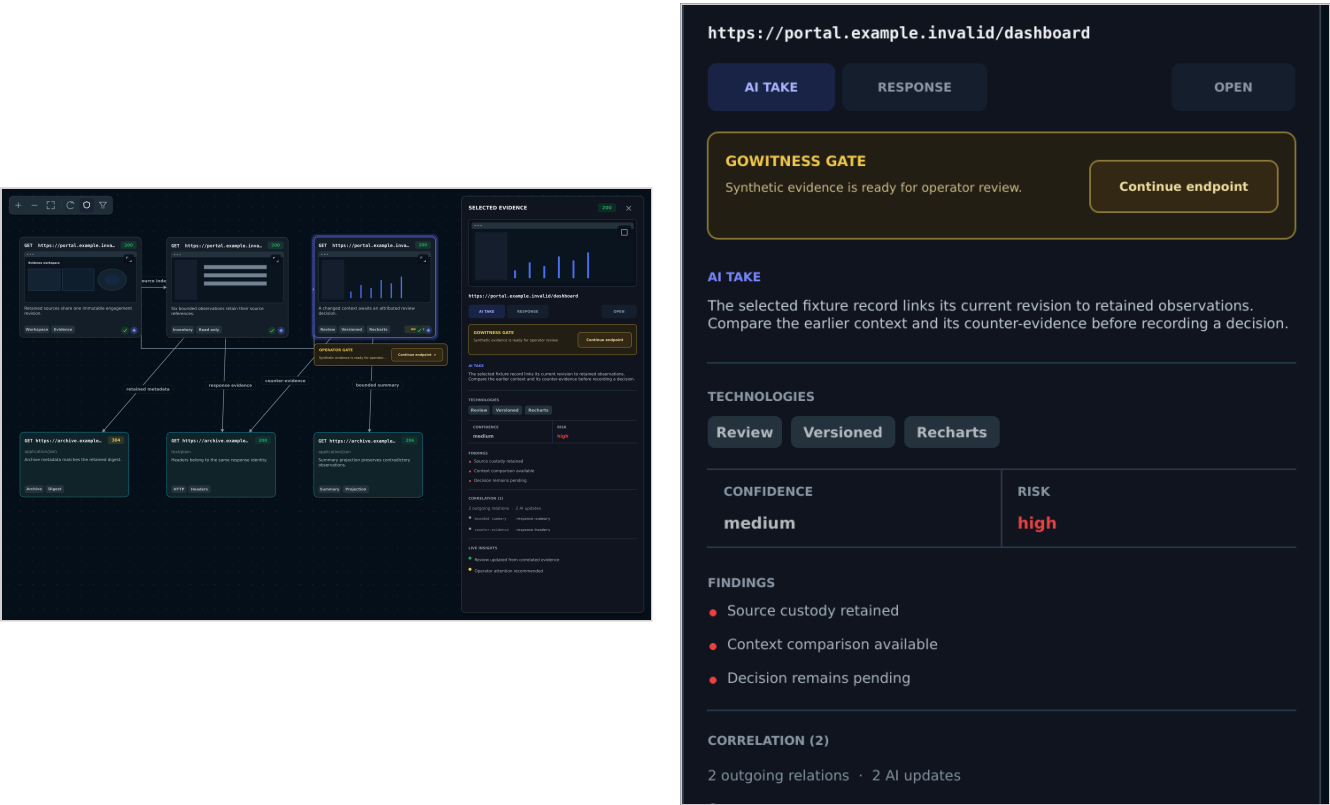
Reference 28

canonical specialist contracts

WEBRECON REVIEW AND GATE STATE

A pending decision has a scope

A gate should explain the affected endpoint or branch and retain the evidence needed for a decision. The rest of the run should not appear globally blocked unless it was explicitly paused.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data. Overview and detail crop from identical source pixels.

The displayed review state identifies the affected endpoint and the evidence associated with the pending decision.

Native interface

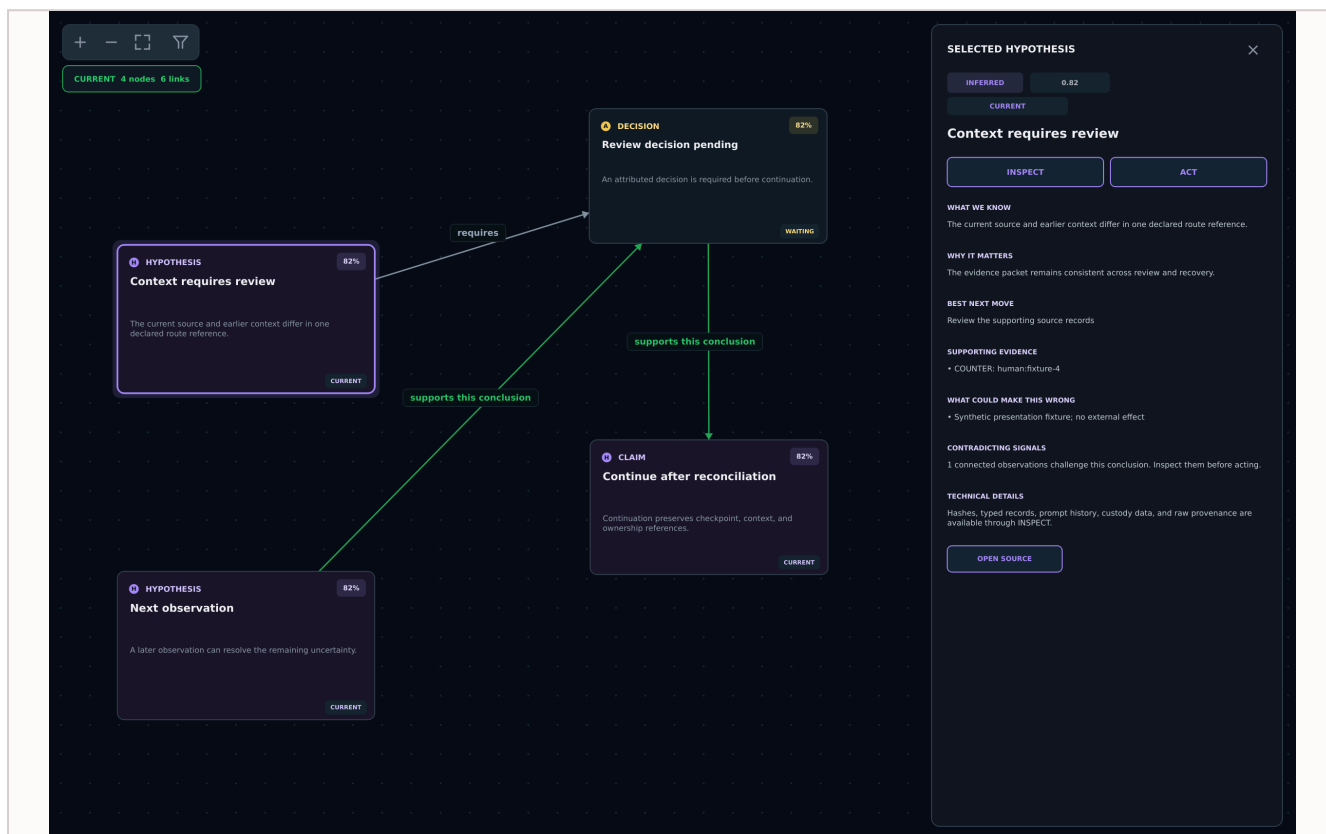
Reference 29

execution graph lifecycle

INVESTIGATION GRAPH

Reasoning needs inspectable sources

Investigation Graph organizes relationships among observations and interpretations. Its companion detail views preserve source references so the operator can inspect the basis of a hypothesis.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

The relationship view and detail panel expose the observations behind a hypothesis without merging them into the interpretation itself.

Native interface

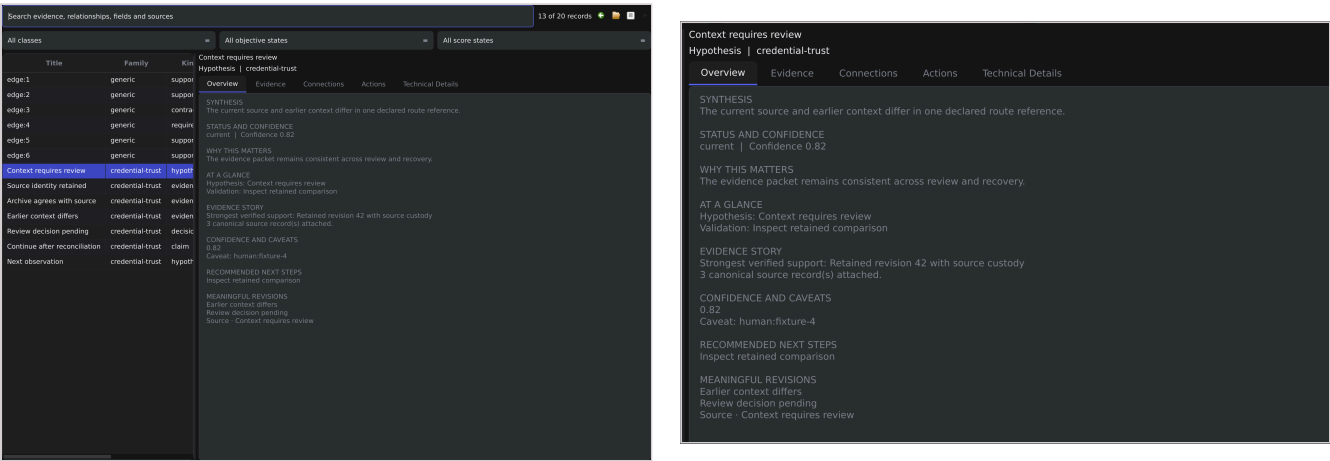
Reference 30

investigation attention

INVESTIGATION WORKBENCH

The workbench exposes evidence and interpretation details

The Workbench provides a more deliberate place to inspect evidence, context and available continuation. Dense records benefit from selection and detail rather than forcing every field into the graph itself.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data. Overview and detail crop from identical source pixels.

Selected records expose their source references and available context in a dedicated detail surface.

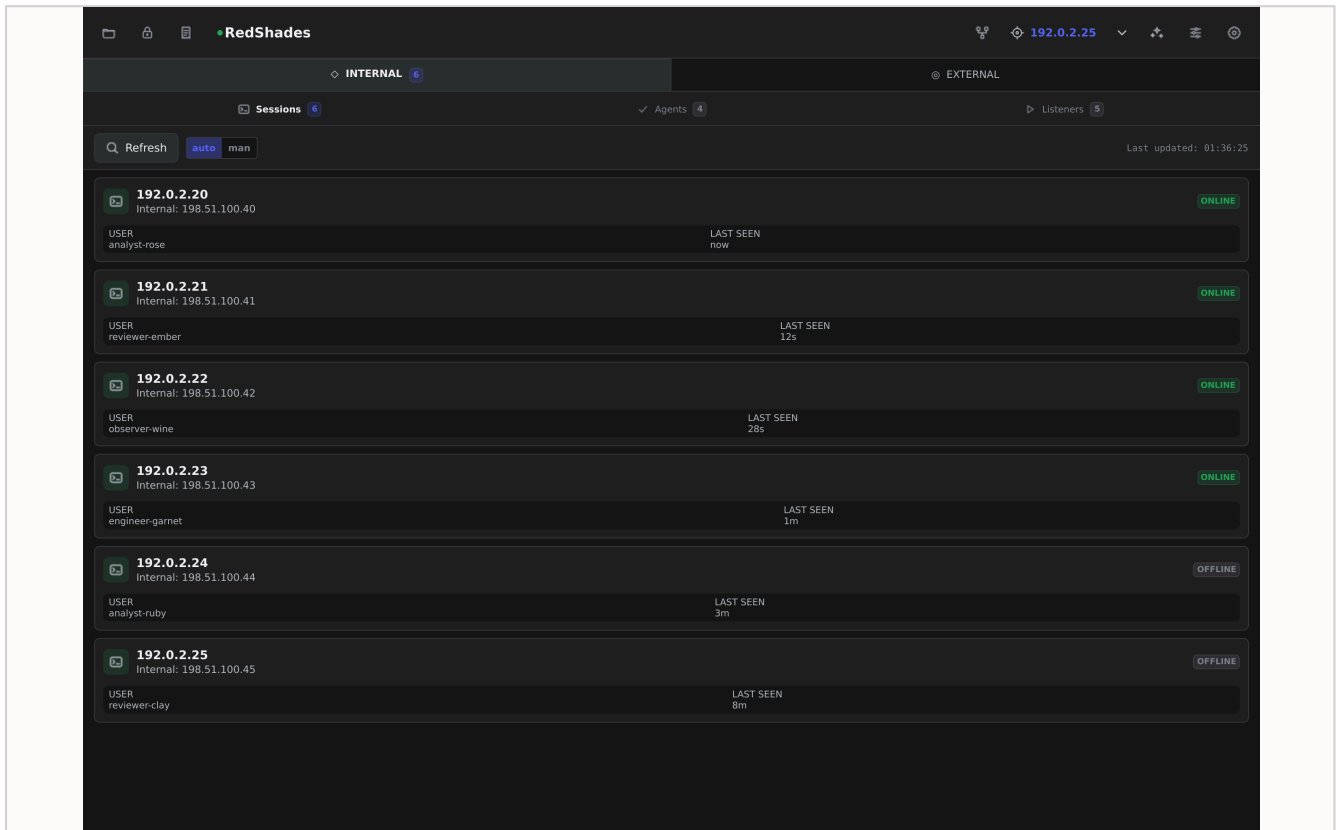
Native interface

Reference 31
canonical specialist contracts

SESSION AND ARTIFACT SURFACES

Sessions and artifacts at a glance

Session and agent views bring stored records, versioned builds and observed availability into the operator workspace. Selected detail connects the inventory with its context and current state.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

The session inventory and selected record provide complementary views of current workspace state.

Native interface

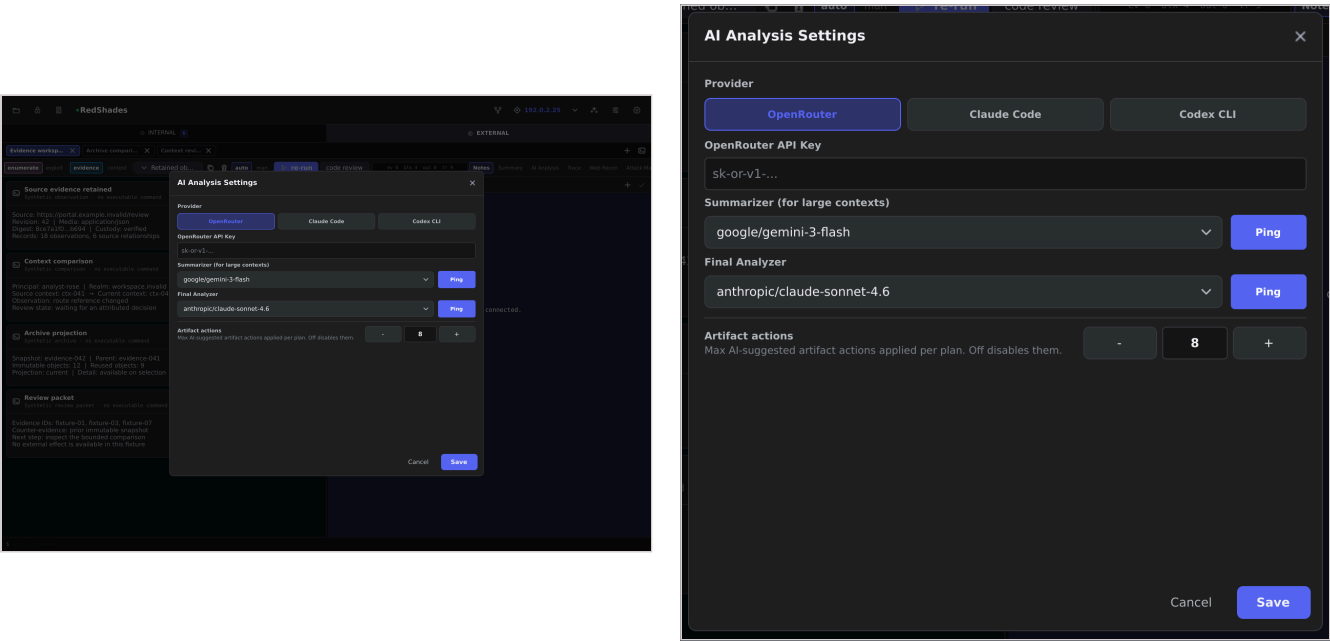
Reference 32

subsystem glossary

MODEL AND RUNTIME CONTROLS

Model and runtime controls in reach

Operator settings expose model and runtime choices in the same workspace as the investigation. Configuration surfaces connect human intent with backend admission, attribution and accounting.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data. Overview and detail crop from identical source pixels.

Model selection and runtime settings remain accessible alongside the investigation interface.

Native interface

Reference 33

authoritative engagement architecture

VERIFICATION COVERAGE

Verification covers components and integrated workflows

Focused contracts, integrated simulations, race checks, repetition, fuzzing and fault injection exercise different dimensions of the platform. Mega and Ultra organize these into explicit engineering acceptance work.

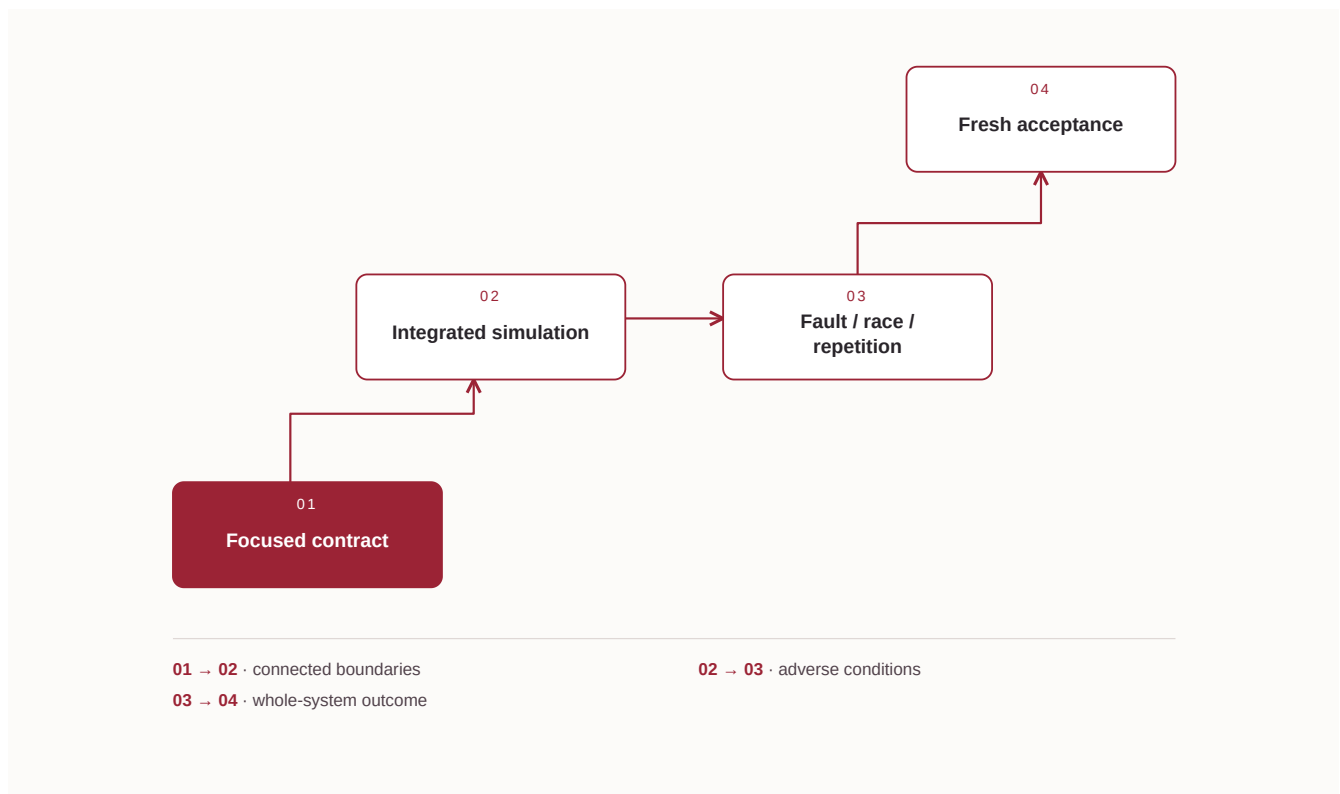


Diagram of the labeled component relationships.

Revision-bound records connect each check to its command and output. Validation status is documented separately from the product's architectural capabilities.

Implemented

Reference 34

ultra ci requirements

DISCLOSURE STUDY

Disclosures become a reusable methodology vocabulary

The Trigger Methodology Graph study organized 200 public disclosures into 37 recurring families. In a 40-disclosure identity-disjoint holdout, 39 belonged to a family represented in that vocabulary.

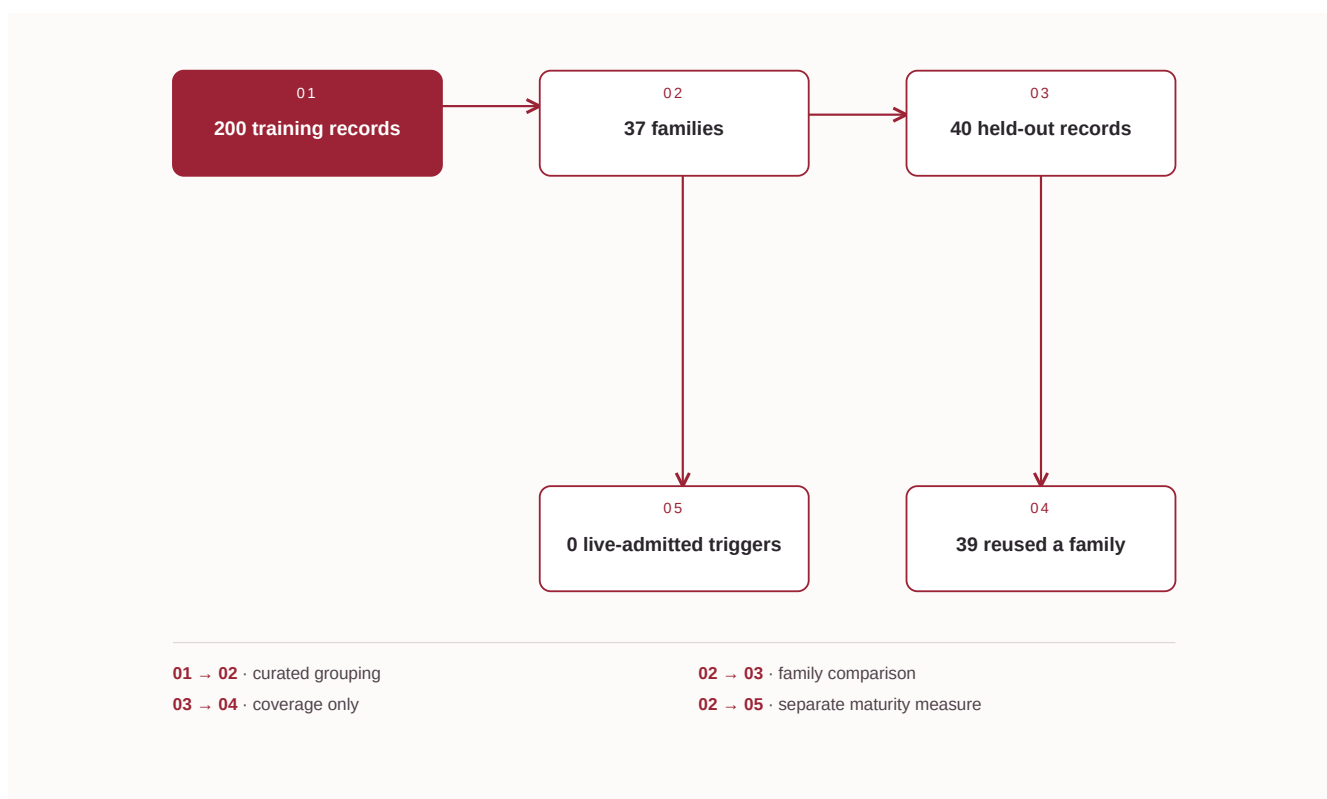


Diagram of the labeled component relationships.

This is family coverage in a curated study, not a count of executed discoveries. It motivates evidence-bound hypotheses and controlled follow-up through the diagnostic graph.

Retained study

Reference 35
tmg disclosure generalization study

EVIDENCE-GAP ANALYSIS

A hypothesis should reveal its missing evidence

Trigger Methodology Graph (TMG) produces evidence-bound diagnostic hypotheses. Shadow Capability Graph (SCG) reasons about missing prerequisites from an immutable snapshot. Bounty Hunting Gates (BHG) retain policy review for proposed hypotheses.

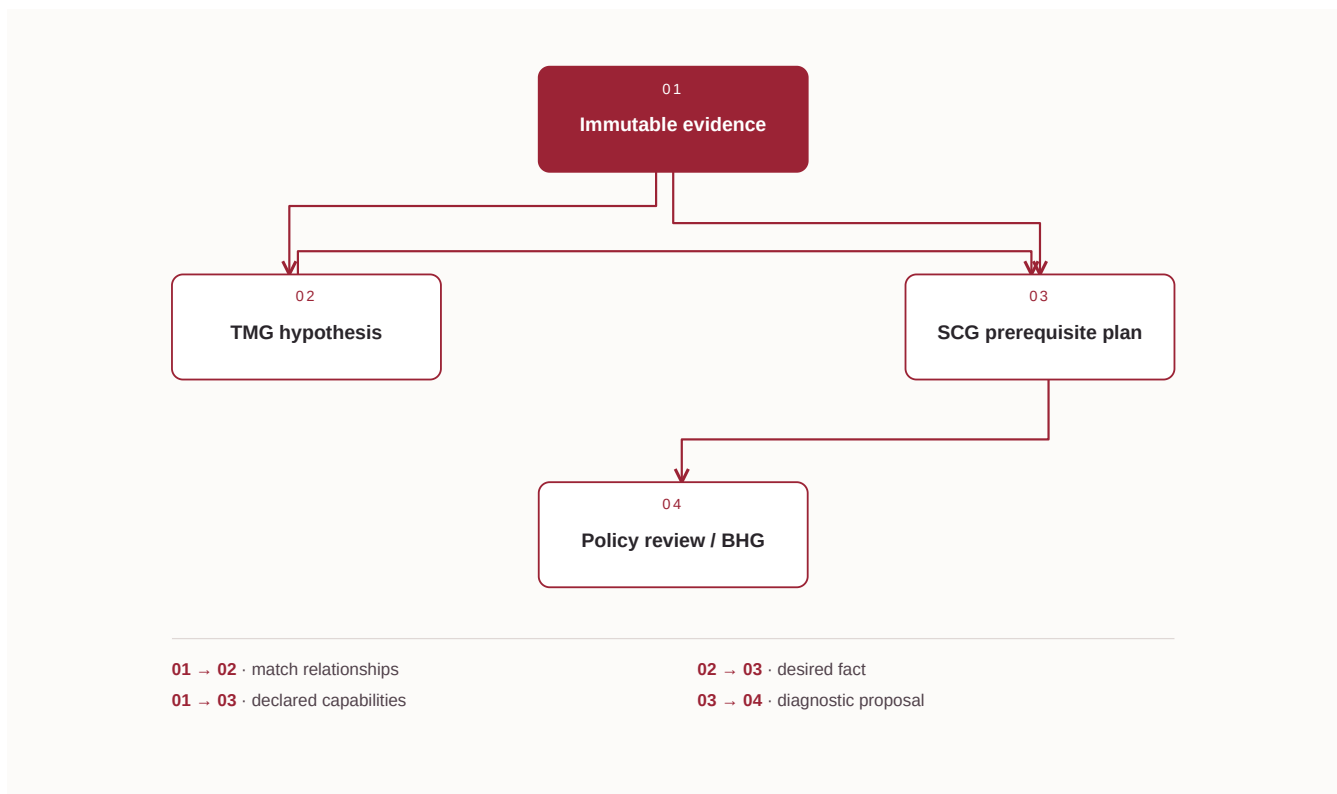


Diagram of the labeled component relationships.

Their useful role is to make uncertainty and the next required evidence explicit. A diagnostic graph does not become scheduling or execution authority simply because its suggestion is plausible.

Implemented

Reference 36

trigger methodology graph + shadow capability graph

REPORTED OUTCOMES

Autonomous outcomes across Windows and web labs

The developer reports unattended completion of P.O.O, a hard Windows CTF, an insane Windows HTB lab and roughly two easy Linux/web HTB labs.

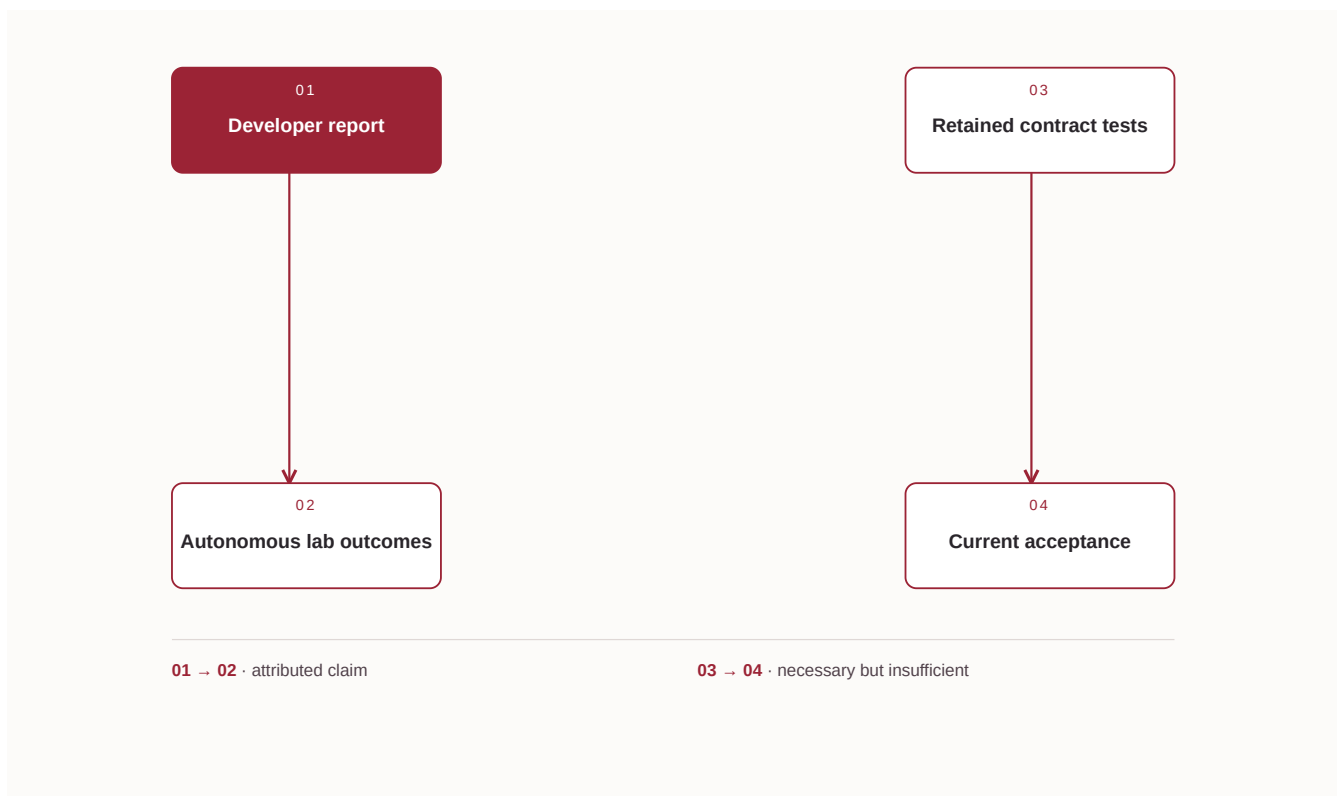


Diagram of the labeled component relationships.

These reported outcomes concern individual environments. They are not a benchmark of performance across unseen targets.

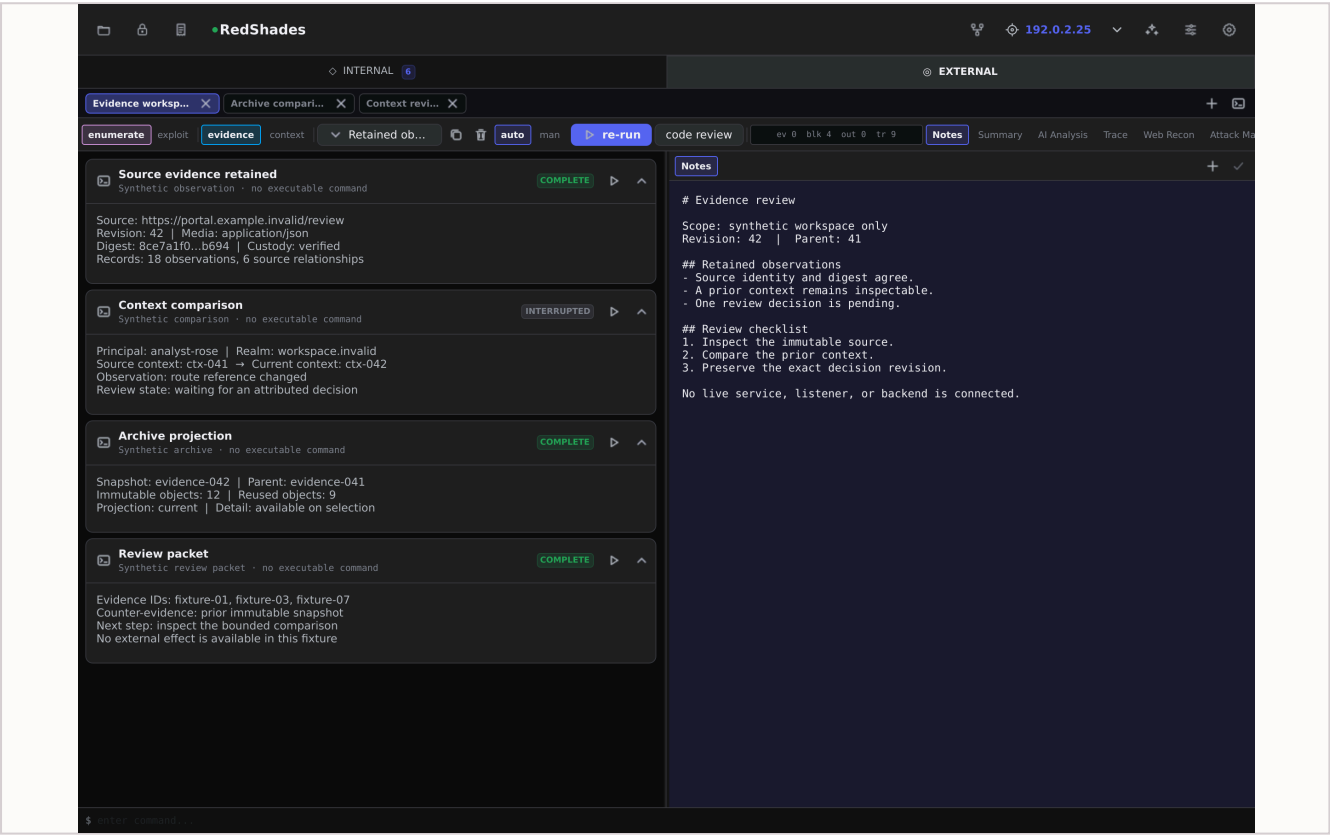
Reported outcomes

Reference 37
design history

TECHNICAL DEMONSTRATION

Technical demonstrations expose the operating model

A private demonstration can examine how the operator workspace connects observed evidence to conditional work, model decisions and reviewed capabilities. The same records show how persistent sessions and runtime replacement preserve an investigation’s context.



Native Qt interface captured on 5 September 2026 with fictional, disconnected demonstration data.

The demonstration covers the native interface and the engineering boundaries described in this publication. Stage scope, model entitlements and annual per-seat pricing are detailed in the bundle comparison.

Product demonstration

Reference 38
editorial thesis

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

implant deployment Validated Core`implant_deployment`

Artifact and session work · Admitted artifact and authorized context → Artifact/session records

ACL operator review (post-summary) Validated Core`acl_operator_action`

Directory analysis · Directory evidence and operator review → Review record

AD compositional capability planning Validated Core`ad_capability_planning`

Directory analysis · Directory evidence → Directory analysis records

AD effective capability discovery Validated Core`ad_capability_discovery`

Directory analysis · Directory evidence → Directory analysis records

BloodHound collection Validated Core`bloodhound_collection`

Directory analysis · Directory context → Graph analysis material

Cross-realm Kerberos service-ticket custody Validated Core`ad_cross_realm_kerberos`

Directory analysis · Directory evidence → Directory analysis records

masscan discovery Validated Core`discovery_masscan`

Discovery · Scope and observation context → Network observations

nmap service and OS scan Validated Core`discovery_nmap`

Discovery · Scope and observation context → Network observations

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

passive OSINT (WHOIS / DNS / ASN) Validated Core`osint_passive`

Discovery • Scope and public observations → Context evidence

query discovered ports Validated Core`discovery_ports`

Discovery • Scope and observation context → Network observations

BloodHound ACL attack path analysis Validated Core`post_acl_analysis`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

chisel SOCKS5 reverse pivot Validated Core`post_pivot_chisel`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

DCSync domain hash dump Validated Core`post_dcsync`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

Disabled legacy advisory handoff Validated Core`post_nopac`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

DPAPI artifact analysis (operator action) Validated Core`post_dpapi`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

Final assessment report Validated Core`post_pwndoc_export`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

Linux privilege escalation enumeration Validated Core`post_linux_privesc`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

pass-the-hash sweep Validated Core`post_pth`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

pass-the-ticket (golden/silver ticket) Validated Core`post_ptt`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

secretsdump credential harvest Validated Core`post_secretsdump`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

SMB binary credential extraction Validated Core`post_smb_binary_analysis`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

Windows privilege escalation enumeration Validated Core`post_windows_privesc`

Extended investigation • Session, artifact or investigation context → Attributed investigation records

AD coercion (PrinterBug/PetitPotam/DFSCoerce) Validated Core`identity_coerce`

Identity analysis • Identity and service context → Identity evidence

certipy AD CS Validated Core`identity_certipy`

Identity analysis • Identity and service context → Identity evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

certipy AD CS exploitation (ESC1/ESC4/ESC8) Validated Core

identity_certipy_exploit

Identity analysis • Identity and service context → Identity evidence

enum4linux-ng Validated Core

identity_enum4linux

Identity analysis • Identity and service context → Identity evidence

GetNPUsers Validated Core

identity_getnpusers

Identity analysis • Identity and service context → Identity evidence

GetUserSPNs Validated Core

identity_getuserspns

Identity analysis • Identity and service context → Identity evidence

hash cracking Validated Core

identity_hash_cracking

Identity analysis • Identity and service context → Identity evidence

kerbrute Validated Core

identity_kerbrute

Identity analysis • Identity and service context → Identity evidence

LAPS password dump Validated Core

identity_laps

Identity analysis • Identity and service context → Identity evidence

ldapdomaindump Validated Core

identity_ldapdomaindump

Identity analysis • Identity and service context → Identity evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

lookupsid Validated Core`identity_lookupsid`**Identity analysis** • Identity and service context → Identity evidence**NTLM relay (ntlmrelayx)** Validated Core`identity_ntlm_relay`**Identity analysis** • Identity and service context → Identity evidence**nxc null/guest enumeration** Validated Core`identity_nxc_guest`**Identity analysis** • Identity and service context → Identity evidence**Responder NTLM hash capture** Validated Core`identity_ntlm_capture`**Identity analysis** • Identity and service context → Identity evidence**targeted credential spray** Validated Core`identity_credential_spray`**Identity analysis** • Identity and service context → Identity evidence**username permutation generation** Validated Core`identity_username_anarchy`**Identity analysis** • Identity and service context → Identity evidence**credential reuse sweep** Validated Core`credential_reuse_sweep`**Investigation** • Scoped evidence → Attributed records**Execution context: network path and domain time** Validated Core`environment_adaptation`**Platform adaptation** • Environment context → Adaptation records

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

conditional mssqlclient Validated Core`remote_mssqlclient`

Remote services · Service and authorized session context → Protocol evidence

conditional smbclient Validated Core`remote_smbclient`

Remote services · Service and authorized session context → Protocol evidence

Kerberos ticket-backed WinRM validation Validated Core`remote_kerberos_winrm`

Remote services · Service and authorized session context → Protocol evidence

MSSQL linked server ADIDNS poisoning Validated Core`mssql_linked_server_poison`

Remote services · Database context → Database evidence

SMB file harvest Validated Core`remote_smb_harvest`

Remote services · Service and authorized session context → Protocol evidence

smbmap share enumeration Validated Core`remote_smbmap`

Remote services · Service and authorized session context → Protocol evidence

deterministic operator report Validated Core`summary`

Reporting · Retained investigation evidence → Report projection

DNS zone transfer drill Validated Core`service_dns_axfr`

Service analysis · Service observations → Service evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

dnsx Validated Core`service_dnsx`

Service analysis • Service observations → Service evidence

FTP anonymous Validated Core`service_ftp_anon`

Service analysis • Service observations → Service evidence

IPMI enumeration and hash extraction Validated Core`service_ipmi`

Service analysis • Service observations → Service evidence

MySQL probe Validated Core`service_mysql`

Service analysis • Service observations → Service evidence

NFS enumeration Validated Core`service_nfs`

Service analysis • Service observations → Service evidence

Redis enumeration Validated Core`service_redis`

Service analysis • Service observations → Service evidence

service CVE candidate sweep Validated Core`service_searchsploit_sweep`

Service analysis • Service observations → Service evidence

service CVE workbench Validated Core`service_cve_workbench`

Service analysis • Service observations → Service evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

SMTP user enumeration Validated Core`service_smtp_user_enum`

Service analysis · Service observations → Service evidence

SNMP enumeration Validated Core`service_snmp_enum`

Service analysis · Service observations → Service evidence

ssh-audit Validated Core`service_ssh_audit`

Service analysis · Service observations → Service evidence

subfinder Validated Core`service_subfinder`

Service analysis · Service observations → Service evidence

credential to shell attempts Validated Core`shell_attempts`

Session work · Authorized execution context → Session-related records

xfreerdp RDP session Validated Core`rdp_session`

Session work · Authorized session context → Session records

CMS scanning Validated Core`web_cms_scan`

Web investigation · Web observations and scoped review → Web evidence

CMS/app-server RCE (Tomcat/Jenkins/Drupal) Validated Core`web_cms_exploit`

Web investigation · Web observations and scoped review → Web evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

CVE PoC workbench review Validated Core`web_cve_poc_workbench`**Web investigation** · Web observations and scoped review → Web evidence**GoWitness endpoint review** Validated Core`web_screenshot`**Web investigation** · Web observations and scoped review → Web evidence**HTTP verb tampering** Validated Core`web_verb_tampering`**Web investigation** · Web observations and scoped review → Web evidence**IDOR / parameter enumeration** Validated Core`web_idor_probe`**Web investigation** · Web observations and scoped review → Web evidence**LFI exploitation** Validated Core`web_lfi_exploit`**Web investigation** · Web observations and scoped review → Web evidence**nuclei parallel** Validated Core`web_nuclei`**Web investigation** · Web observations and scoped review → Web evidence**OS command injection probe** Validated Core`web_cmdi_probe`**Web investigation** · Web observations and scoped review → Web evidence**post-auth CVE PoC workbench review** Validated Core`web_post_auth_poc_workbench`**Web investigation** · Web observations and scoped review → Web evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

post-auth exploit probes Validated Core

web_post_auth_exploit_probes

Web investigation · Web observations and scoped review → Web evidence

post-auth surface harvest Validated Core

web_post_auth_harvest

Web investigation · Web observations and scoped review → Web evidence

SQLi OS shell (--os-cmd / --os-shell) Validated Core

web_sql_i_post_exploit

Web investigation · Web observations and scoped review → Web evidence

SQLi probe Validated Core

web_sql_i_probe

Web investigation · Web observations and scoped review → Web evidence

SSTI probe Validated Core

web_ssti_probe

Web investigation · Web observations and scoped review → Web evidence

target-specific password generation Validated Core

web_cewl

Web investigation · Web observations and scoped review → Web evidence

web content analysis Validated Core

web_content_analysis

Web investigation · Web observations and scoped review → Web evidence

web file upload attack Validated Core

web_file_upload_attack

Web investigation · Web observations and scoped review → Web evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

88 AUTOMATIC DECLARATIONS • CONDITIONAL GRAPH

Core stage atlas

Each entry identifies available work and its evidence context. Separate C2 families extend session work.

web fingerprint + wafw00f Validated Core

web_whatweb_waf

Web investigation · Web observations and scoped review → Web evidence

web fuzzing (vhost+dir+ext) Validated Core

web_fuzz

Web investigation · Web observations and scoped review → Web evidence

web login brute force Validated Core

web_login_bruteforce

Web investigation · Web observations and scoped review → Web evidence

web RCE candidate correlation Validated Core

web_rce_candidate_correlation

Web investigation · Web observations and scoped review → Web evidence

web RCE confirmation Validated Core

web_rce_confirm

Web investigation · Web observations and scoped review → Web evidence

web RCE deployment preparation Validated Core

web_rce_deployment_prepare

Web investigation · Web observations and scoped review → Web evidence

XSS probe Validated Core

web_xss_probe

Web investigation · Web observations and scoped review → Web evidence

XXE probe Validated Core

web_xxe_probe

Web investigation · Web observations and scoped review → Web evidence

Results return to the evidence-driven scheduler. Stage declarations describe available work, not a mandatory linear sequence.

Source-derived inventory

88 Core stages

13 CONTINUATION STAGES

Session work follows platform-specific branches

Registered session context selects Windows, Linux and shared continuation work. These 13 stages complement the 88-stage Validated Core and are included in the 198-stage catalog.

permission enumeration windows windows_permission_enum	Linux permission enumeration linux linux_permission_enum
pivot tunnel + internal discovery Cross-platform pivot_tunnel	filesystem enumeration windows windows_filesystem_enum
expanded privilege escalation enumeration windows windows_privilege_escalation	privilege escalation exploitation windows windows_privilege_escalation_exploit
Windows lateral movement windows windows_lateral_movement	Windows persistence windows windows_persistence
Linux filesystem enumeration linux linux_filesystem_enum	expanded Linux privilege escalation enumeration linux linux_privilege_escalation
Linux privilege escalation exploitation linux linux_privilege_escalation_exploit	Linux lateral movement linux linux_lateral_movement
Linux persistence linux linux_persistence	

88 Core + 13 C2 continuation + 80 Advanced + 17 Ultimate = 198 stages.

C2 continuation

PLANNED INTEGRATIONS · PACKAGING PENDING

Specialist domain and tool relationships

48 descriptors: 37 potential third-party packages, one external Apple toolchain and ten native or custom components.
Domain totals overlap through reuse.

Cluster → capability → shared tool architecture

Domain cluster	Analysis capabilities	Primary and alternative components
Android / mobile 8 stages · 10 referenced entries	artifact intake · provenance custody · configuration review · static analysis · topology modeling · evidence correlation · emulation simulation · reporting	Androguard · Apktool · Android bundletool · Cosign · JADX · Mobile Security Framework · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades evidence-backed reporting
Apple platforms 8 stages · 8 referenced entries	artifact intake · provenance custody · binary decomposition · configuration review · static analysis · topology modeling · emulation simulation · reporting	ipsw · LLVM binary utilities · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades privacy/data-flow model · RedShades evidence-backed reporting · Apple codesign/security toolchain
Blockchain 8 stages · 8 referenced entries	artifact intake · provenance custody · static analysis · evidence correlation · emulation simulation · topology modeling · reporting	Cosign · Echidna · Foundry · RedShades artifact intake · RedShades typed evidence graph · RedShades evidence-backed reporting · Slither · Solidity compiler
Bluetooth / BLE / NFC 7 stages · 8 referenced entries	artifact intake · passive capture parsing · configuration review · evidence correlation · topology modeling · reporting	BlueZ btmon · libnfc utilities · RedShades artifact intake · RedShades typed evidence graph · RedShades protocol observation normalizer · RedShades evidence-backed reporting · Wireshark/TShark · Zeek
Wi-Fi / RF 8 stages · 9 referenced entries	artifact intake · topology modeling · passive capture parsing · static analysis · evidence correlation · policy gating · reporting	GNU Radio · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades protocol observation normalizer · RedShades evidence-backed reporting · Suricata · Wireshark/TShark · Zeek
LLM / AI systems 8 stages · 10 referenced entries	artifact intake · topology modeling · configuration review · emulation simulation · evidence correlation · supply chain analysis · reporting	garak · Inspect AI · OSV-Scanner · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades privacy/data-flow model · RedShades evidence-backed reporting · Syft · Trivy

97 expansion stages: 80 Advanced (78 domain stages and two shared helpers) and 17 Ultimate. Licensing and validation apply to each integration. The 48 entries include 47 stage-referenced components and one foundational router.

Under testing / Under development

22 open-source integration candidates · 15 license-review candidates · 5 built-in · 5 custom planned · 1 external

PLANNED INTEGRATIONS • PACKAGING PENDING

Specialist domain and tool relationships

48 descriptors: 37 potential third-party packages, one external Apple toolchain and ten native or custom components.
Domain totals overlap through reuse.

Cluster → capability → shared tool architecture

Domain cluster	Analysis capabilities	Primary and alternative components
Satellite / space 7 stages · 10 referenced entries	artifact intake · passive capture parsing · configuration review · topology modeling · emulation simulation · reporting	GNU Radio · gr-satellites · Orekit · RedShades artifact intake · RedShades typed evidence graph · RedShades protocol observation normalizer · RedShades evidence-backed reporting · RedShades OT/space safety model · Wireshark/TShark · Zeek
Quantum readiness 6 stages · 6 referenced entries	cryptographic inventory · evidence correlation · configuration review · topology modeling · emulation simulation · reporting	Cirq · Qiskit SDK · RedShades artifact intake · RedShades cryptographic evidence normalizer · RedShades typed evidence graph · RedShades evidence-backed reporting
Hardware / firmware 9 stages · 11 referenced entries	artifact intake · binary decomposition · supply chain analysis · static analysis · configuration review · evidence correlation · emulation simulation · reporting	Binwalk · Cosign · Ghidra · ipsw · LLVM binary utilities · QEMU · RedShades artifact intake · RedShades typed evidence graph · RedShades evidence-backed reporting · Syft · Trivy
ICS / SCADA 9 stages · 11 referenced entries	artifact intake · passive capture parsing · topology modeling · safety modeling · configuration review · static analysis · evidence correlation · emulation simulation · policy gating	OpenSCAP · QEMU · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades protocol observation normalizer · RedShades evidence-backed reporting · RedShades OT/space safety model · Suricata · Wireshark/TShark · Zeek
Defensive patching 8 stages · 9 referenced entries	supply chain analysis · evidence correlation · static analysis · reporting · policy gating	Cosign · Grype · OSV-Scanner · RedShades typed evidence graph · RedShades defensive change planner · RedShades policy and effect gate · RedShades evidence-backed reporting · Syft · Trivy
Digital forensics 9 stages · 12 referenced entries	provenance custody · artifact intake · forensic analysis · evidence correlation · reporting	Cosign · Plaso · RedShades artifact intake · RedShades typed evidence graph · RedShades policy and effect gate · RedShades evidence-backed reporting · The Sleuth Kit · Velociraptor · Volatility 3 · Wireshark/TShark · YARA · Zeek
Shared Advanced helpers 2 stages · 3 referenced entries	evidence correlation	RedShades artifact intake · RedShades typed evidence graph · RedShades evidence-backed reporting

The two shared helpers normalize cross-domain evidence and correlate observations into a converged assessment record.

97 expansion stages: 80 Advanced (78 domain stages and two shared helpers) and 17 Ultimate. Licensing and validation apply to each integration. The 48 entries include 47 stage-referenced components and one foundational router.

Under testing / Under development

22 open-source integration candidates · 15 license-review candidates · 5 built-in · 5 custom planned · 1 external

UNDER TESTING

Specialist application domains

Android / mobile

8 stages

Under testing

Entry evidence

Application and device evidence

Conditional branches

Package, permission and storage branches

Capabilities

Mobile configuration, application metadata and transport review

Shared tool architecture

10 referenced entries - Androguard, Apktool, Android bundletool, Cosign, JADX, Mobile Security Framework. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Apple platforms

8 stages

Under testing

Entry evidence

iOS, iPhone and macOS evidence

Conditional branches

Application, entitlement and platform branches

Capabilities

Signing, entitlements, privacy settings and platform configuration

Shared tool architecture

8 referenced entries - ipsw, LLVM binary utilities, Apple codesign/security toolchain. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Blockchain

8 stages

Under testing

Entry evidence

Contract and transaction evidence

Conditional branches

Contract, dependency and governance branches

Capabilities

Contract metadata, access controls and transaction provenance

Shared tool architecture

8 referenced entries - Cosign, Echidna, Foundry, Slither, Solidity compiler. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

88 Core + 13 C2 continuation + 80 Advanced + 17 Ultimate = 198 stages.

 Under testing

UNDER TESTING

Wireless and AI assessment domains

Proximity / BLE / NFC

7 stages

Under testing

Entry evidence

Captured proximity observations

Conditional branches

Protocol and device-class branches

Capabilities

Advertisement, pairing and NFC record analysis

Shared tool architecture

8 referenced entries · BlueZ btmon, libnfc utilities, Wireshark/TShark, Zeek. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Wi-Fi / RF

8 stages

Under testing

Entry evidence

Wireless configuration and recorded traffic

Conditional branches

Network, radio and authentication branches

Capabilities

Wireless posture, spectrum records and authentication configuration

Shared tool architecture

9 referenced entries · GNU Radio, Suricata, Wireshark/TShark, Zeek. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

LLM / AI systems

8 stages

Under testing

Entry evidence

Model, application and evaluation evidence

Conditional branches

Data, model and application boundary branches

Capabilities

Evaluation artifacts, data handling and permission boundaries

Shared tool architecture

10 referenced entries · garak, Inspect AI, OSV-Scanner, Syft, Trivy. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

88 Core + 13 C2 continuation + 80 Advanced + 17 Ultimate = 198 stages.

Under testing

UNDER TESTING

Satellite systems and quantum readiness

Satellite / space

7 stages

Under testing

Entry evidence

Ground-system and telemetry artifacts

Conditional branches

Ground, link and telemetry branches

Capabilities

Telemetry formats, asset configuration and trust boundaries

Shared tool architecture

10 referenced entries · GNU Radio, gr-satellites, Orekit, Wireshark/TShark, Zeek. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

Quantum readiness

6 stages

Under testing

Entry evidence

Cryptographic asset inventory

Conditional branches

Algorithm, dependency and migration branches

Capabilities

Cryptographic discovery and migration readiness analysis

Shared tool architecture

6 referenced entries · Cirq, Qiskit SDK. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

88 Core + 13 C2 continuation + 80 Advanced + 17 Ultimate = 198 stages.

Under testing

UNDER TESTING

Physical and industrial systems

Hardware / firmware

9 stages

Under testing

Entry evidence

Firmware images and component inventory

Conditional branches

Image, component and boot-policy branches

Capabilities

Firmware metadata, component provenance and secure-boot posture

Shared tool architecture

11 referenced entries · Binwalk, Cosign, Ghidra, ipsw, LLVM binary utilities, QEMU, Syft, Trivy. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

ICS / SCADA

9 stages

Under testing

Entry evidence

Industrial asset and configuration evidence

Conditional branches

Asset, protocol and zone branches

Capabilities

Configuration review, protocol records and segmentation posture

Shared tool architecture

11 referenced entries · OpenSCAP, QEMU, Suricata, Wireshark/TShark, Zeek. Native evidence and reporting components complete the cluster.

Gate & convergence

Scope and evidence checks select eligible analysis. Findings converge into an attributed domain report; missing inputs keep dependent work pending.

88 Core + 13 C2 continuation + 80 Advanced + 17 Ultimate = 198 stages.

Under testing

UNDER DEVELOPMENT

PurpleShades and digital forensics

PurpleShades • Defensive patching

8 stages • Under development

Asset and finding intake opens parallel exposure, dependency and remediation analysis. A change plan brings the branches together with rollback criteria, canary observations and approval requirements.

Advisories → assessment branches → change proposal → rollback / canary criteria → approval → review packet

The workflow assembles a reviewed proposal and defines the evidence required to validate a future change.

Digital forensics

9 stages • Under development

Case intake establishes evidence identity and custody. Artifact, timeline and correlation branches preserve source attribution and converge into a reviewable case record.

Case intake → evidence preservation → analysis branches → correlation → review → case report

Integrity checks precede analysis. Incomplete or conflicting observations remain visible in the final evidence account.

88 Core + 13 C2 continuation + 80 Advanced + 17 Ultimate = 198 stages.

Under development

ANNUAL PRICING PER SEAT

Annual pricing and bundle entitlements

Core

Starting at

\$7,000

/ year per seat

88 Core stages + 13 C2 continuation stages

Validated Core

100 fine-tuned specialist adapters for bounded stateless calls and code-review roles.

Advanced

Starting at

\$14,000

/ year per seat

Core + 80 Advanced stages

Under testing

Fine-tuned 30B3A Flash model for persistent higher-complexity sessions.

Ultimate

Starting at

\$20,000

/ year per seat

Advanced + 17 defensive and forensic stages

Under development

Proprietary multi-encoder neural representation suite for defensive and forensic workloads, with multiple task-specific embedding and representation variants.

Model and adapter weights are provided under NDA. Active subscriptions receive weights and stage updates at least every six months.

198 stages: 88 Validated Core + 13 C2 continuation + 80 Advanced + 17 Ultimate.

88 Core + 13 C2 continuation + 80 Advanced + 17 Ultimate = 198 stages.

Annual pricing per seat